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Lafferty et al.

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(54) **APPARATUS AND A METHOD FOR
BONDING WEBS OF NON-WOVEN PLASTIC
MATERIAL**

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(71) Applicant: **Dukane IAS, LLC**, St. Charles, IL
(US)

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(72) Inventors: **Justin Marshall Lafferty**, Marshfield,
WI (US); **Matthew James Dittrich**,
Chagrin Falls, OH (US); **Paul J. Golko**,
Crystal Lake, IL (US); **Leo Klinstein**,
Glenview, IL (US)

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(73) Assignee: **Dukane IAS, LLC**, St. Charles, IL
(US)

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Primary Examiner — James D Sells

(74) *Attorney, Agent, or Firm* — ICE MILLER LLP;
Justin D. Swindells

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(57) **ABSTRACT**

An apparatus having a first treatment module and a second treatment module, at least one of which includes a welding device for treating one or more workpieces. The apparatus has a controller arranged to: receive a real-time amplitude signal of ultrasonic vibration of at least one of the first treatment module and the second treatment module during a treatment cycle; segment the amplitude signal into a plurality of amplitude segments for the treatment cycle; monitor an amplitude value of each of the plurality of amplitude segments during the treatment cycle; operate a plurality of closed-loop control algorithms to determine a plurality of amplitude adjustment values, each of the plurality of amplitude adjustment values corresponding to a respective one of the plurality of amplitude segments; and apply each amplitude adjustment value to the corresponding respective one of the plurality of amplitude segments during the subsequent treatment cycle in real-time.

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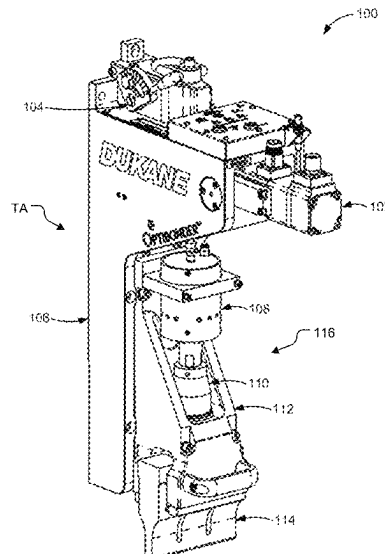
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B29C 65/08 (2006.01)
B29C 65/00 (2006.01)

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(2013.01); **B29C 66/7294** (2013.01)

(58) **Field of Classification Search**
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See application file for complete search history.

20 Claims, 15 Drawing Sheets



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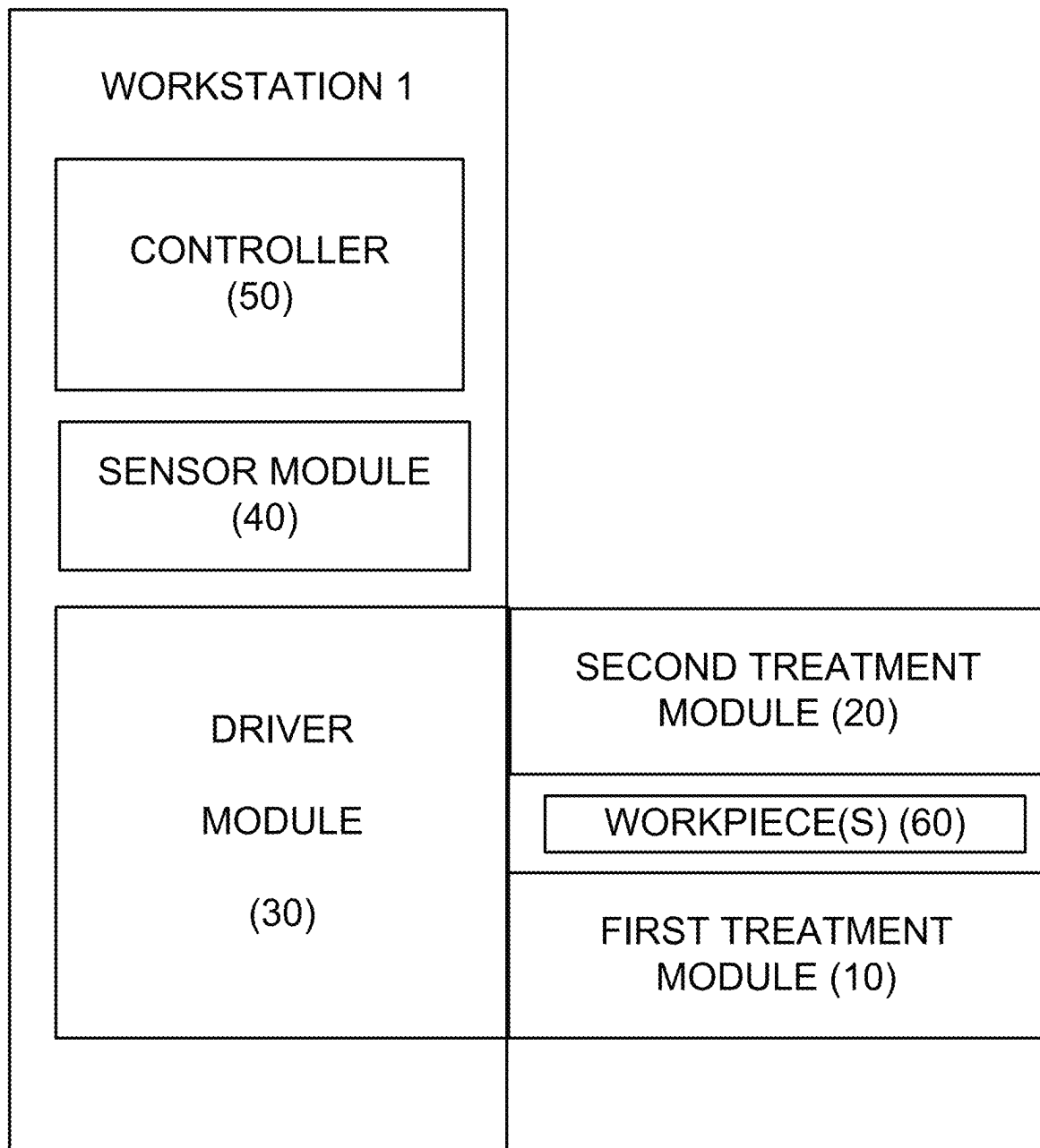


FIG. 1A

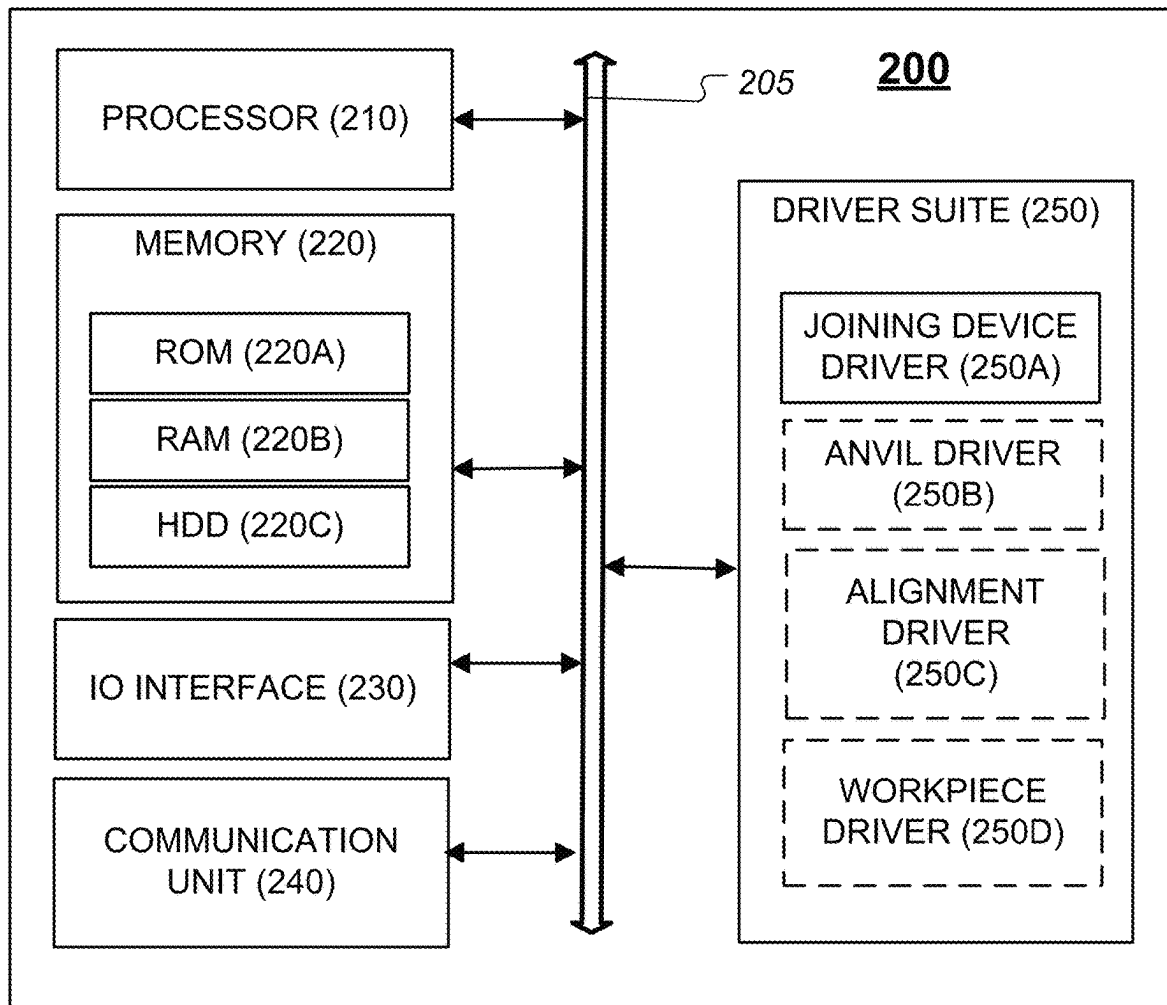


FIG. 1B

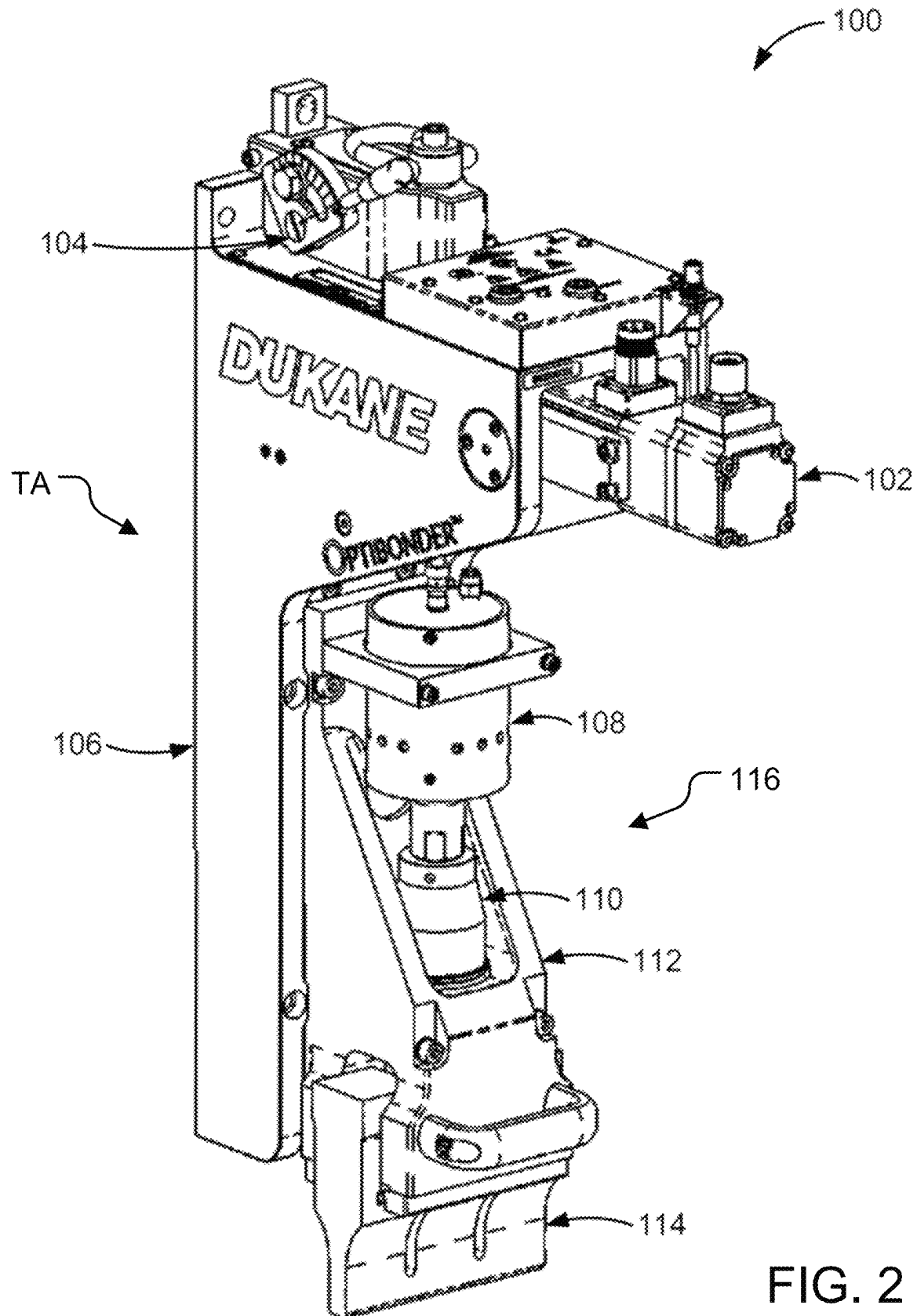


FIG. 2

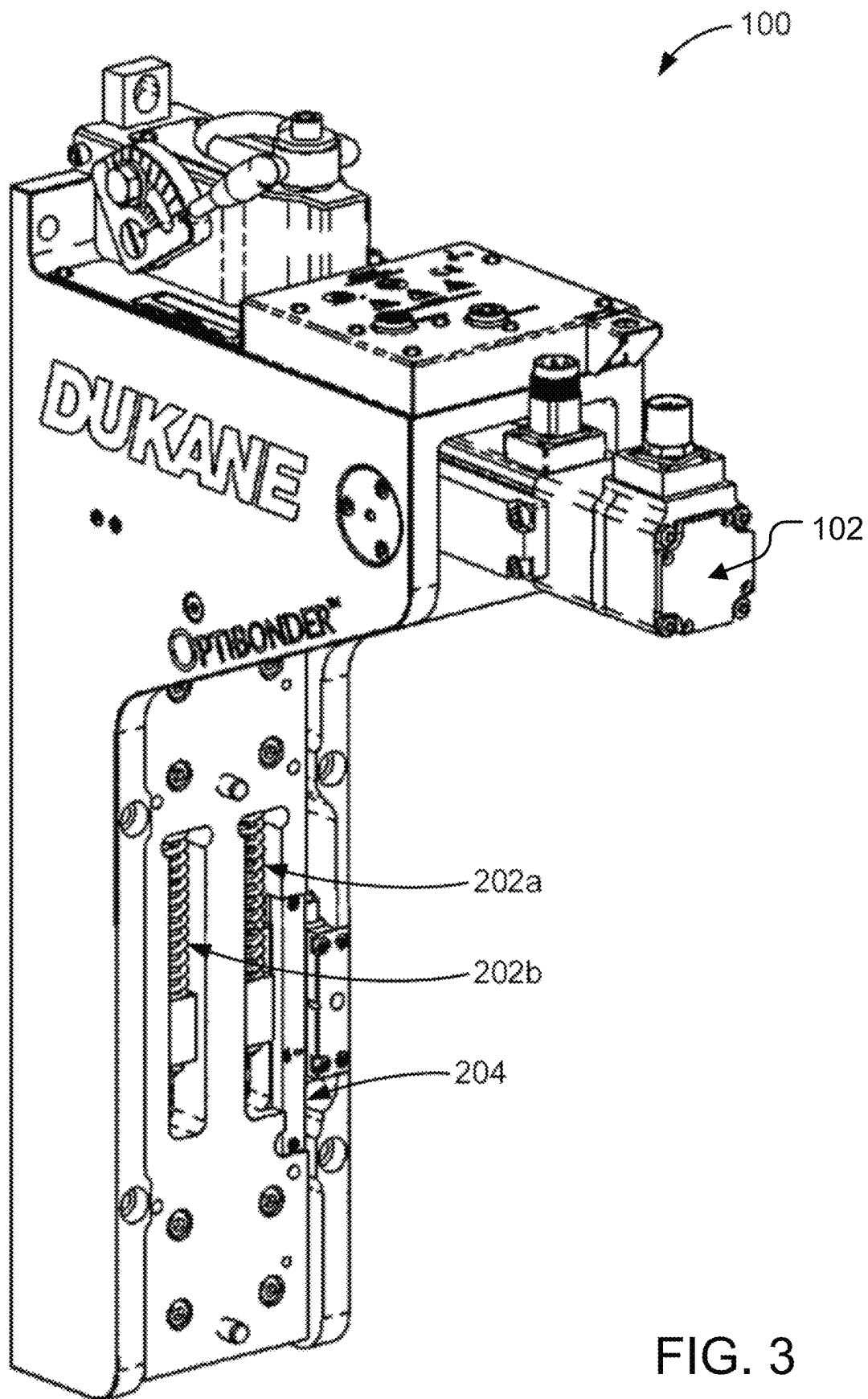
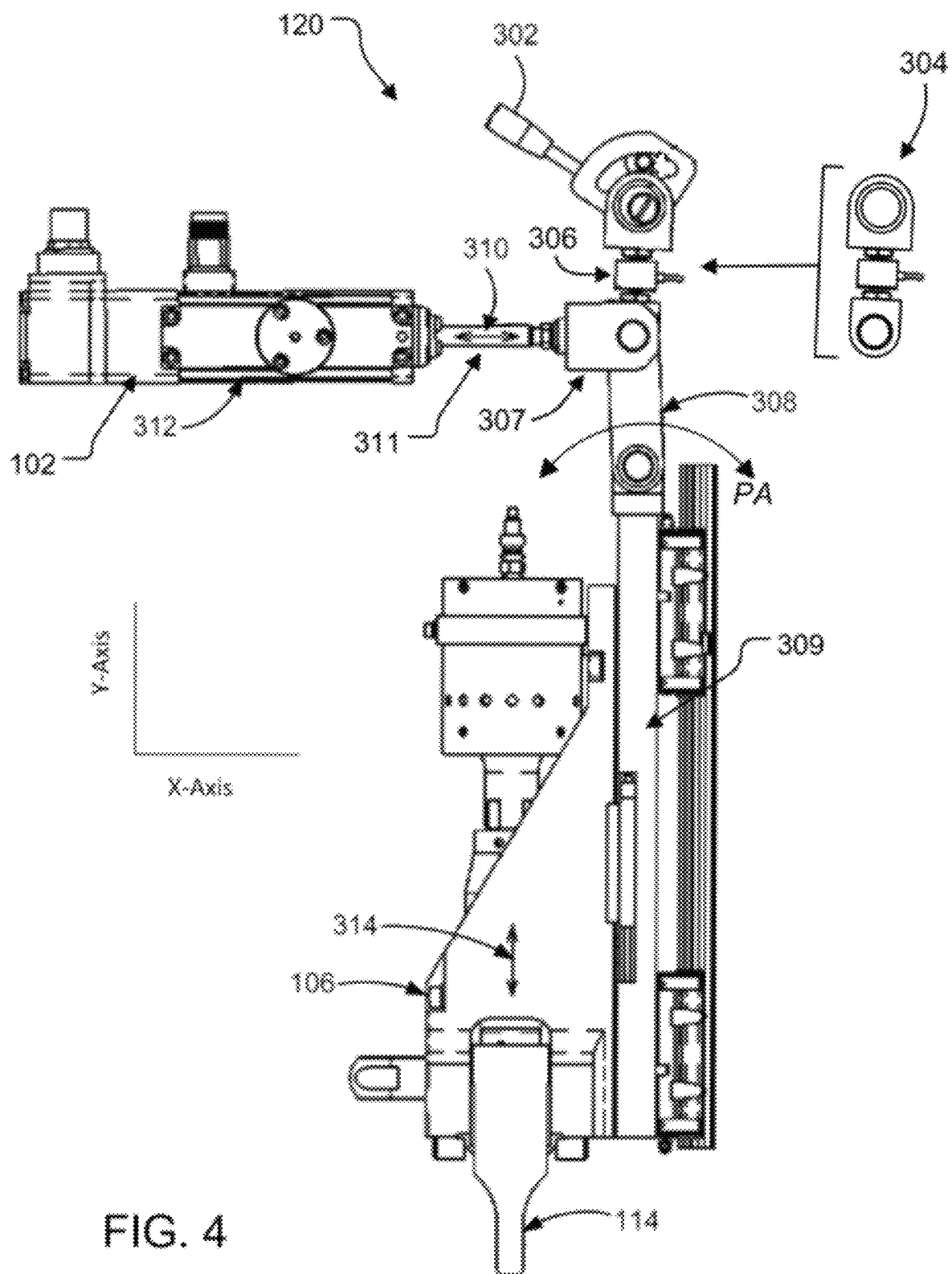


FIG. 3



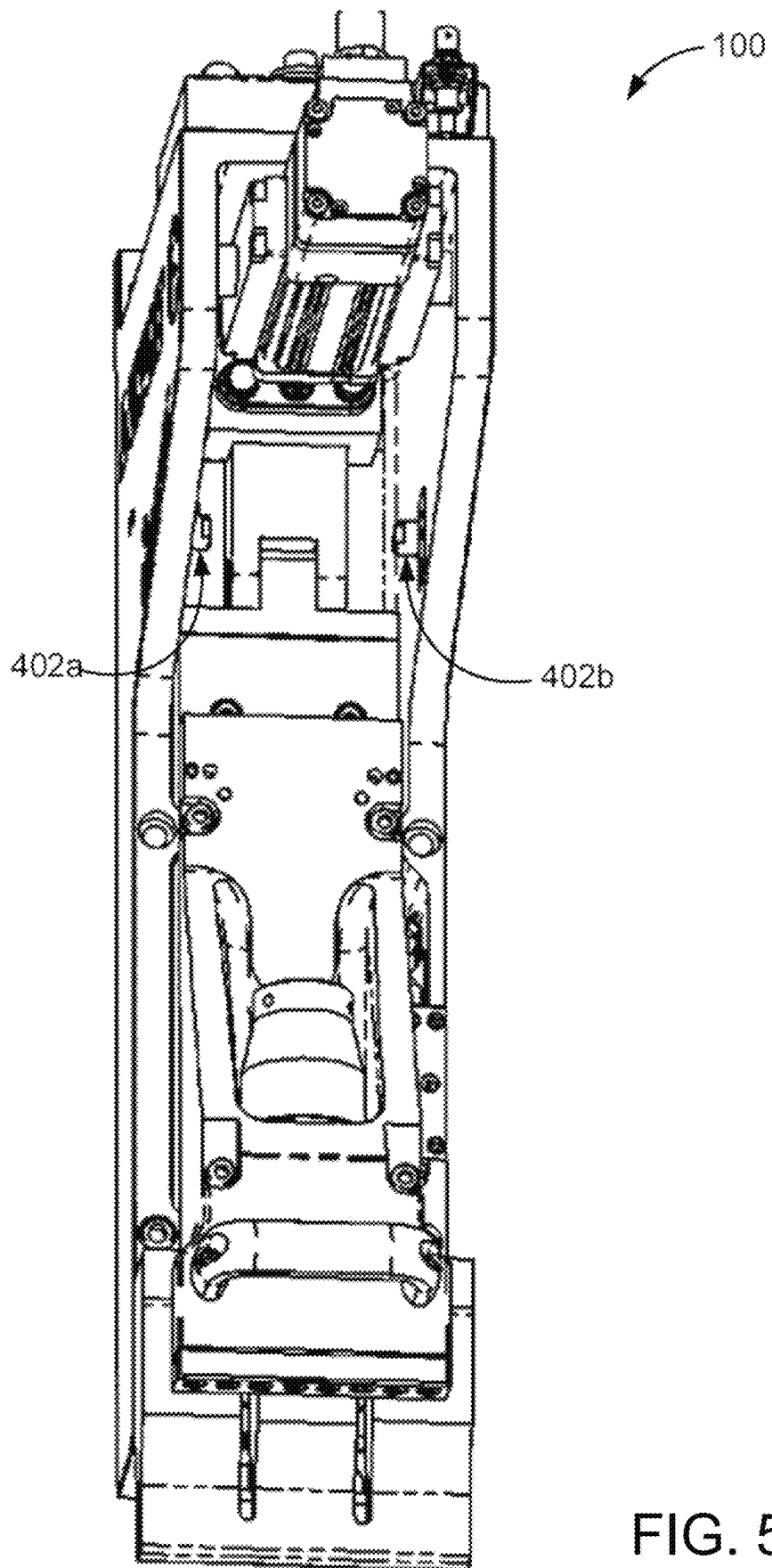


FIG. 5

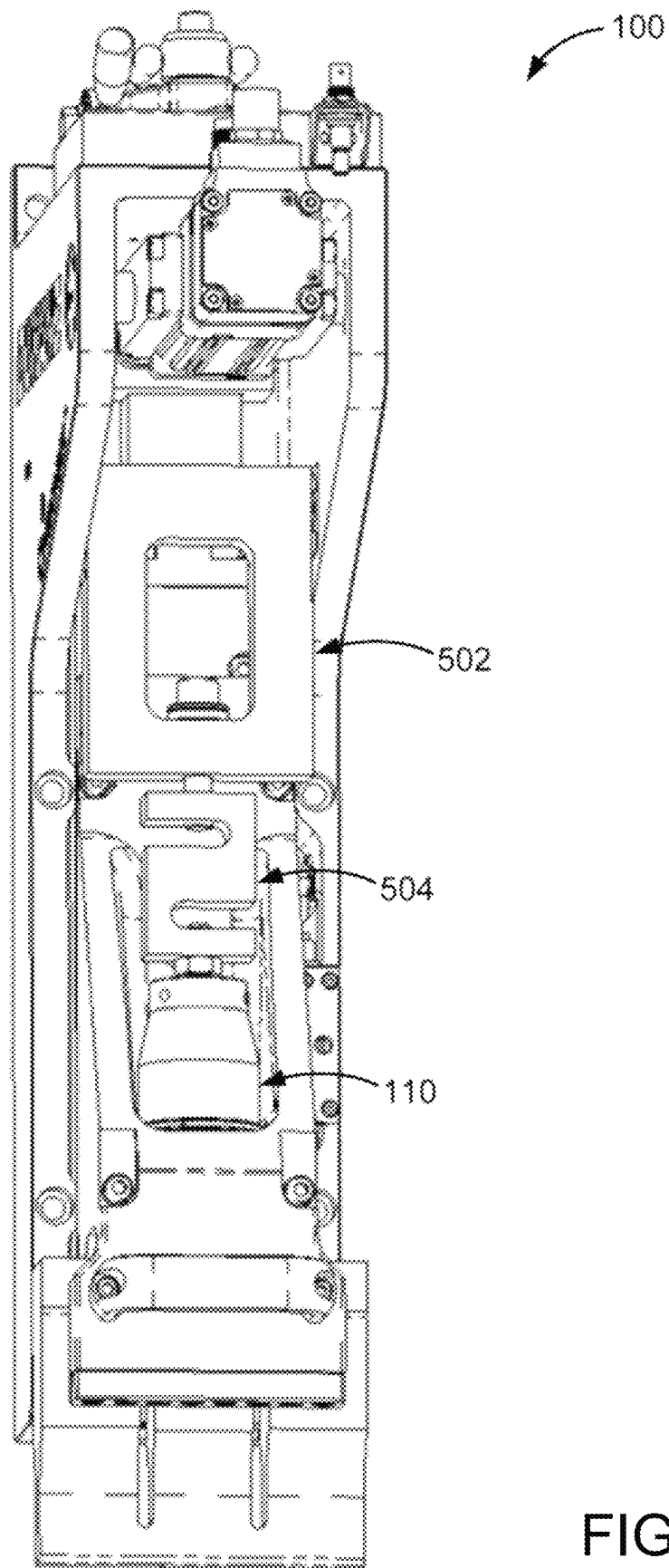
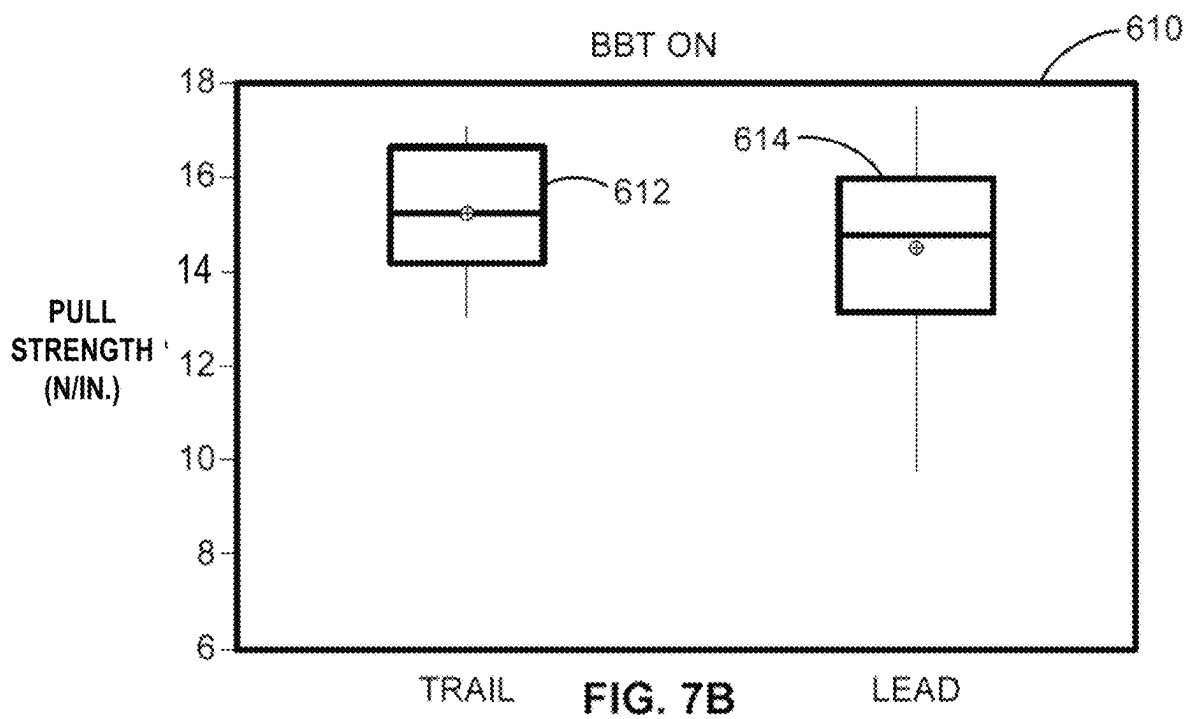
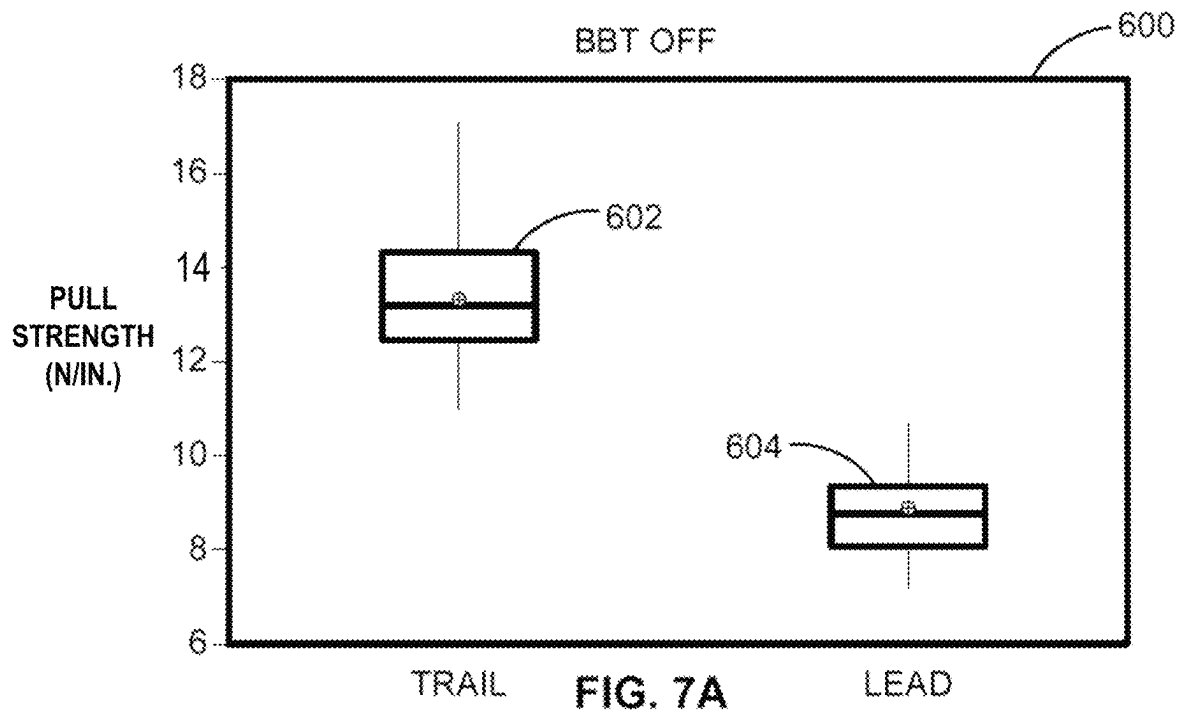


FIG. 6



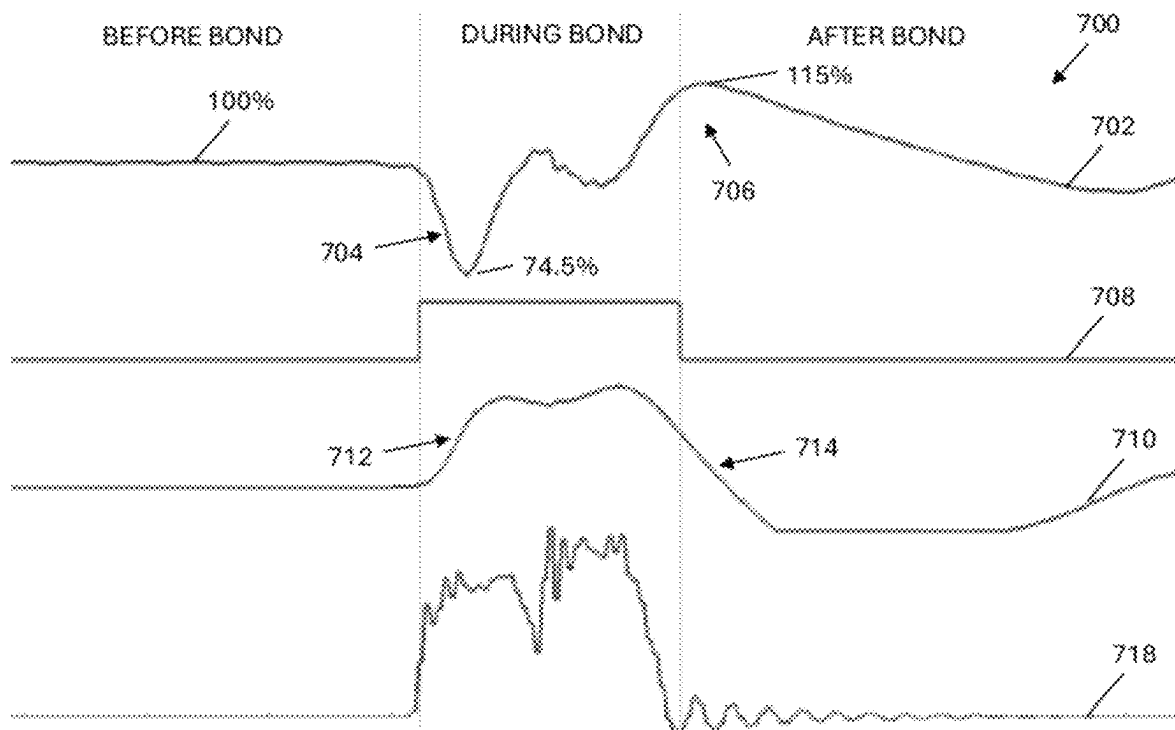


FIG. 8A

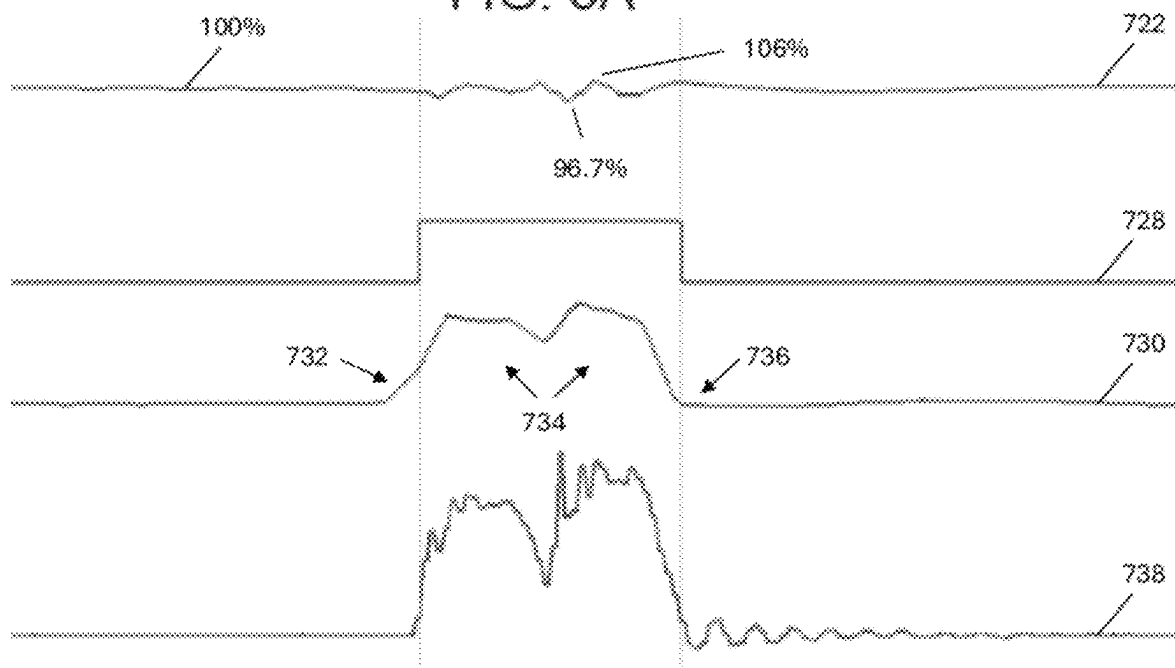


FIG. 8B

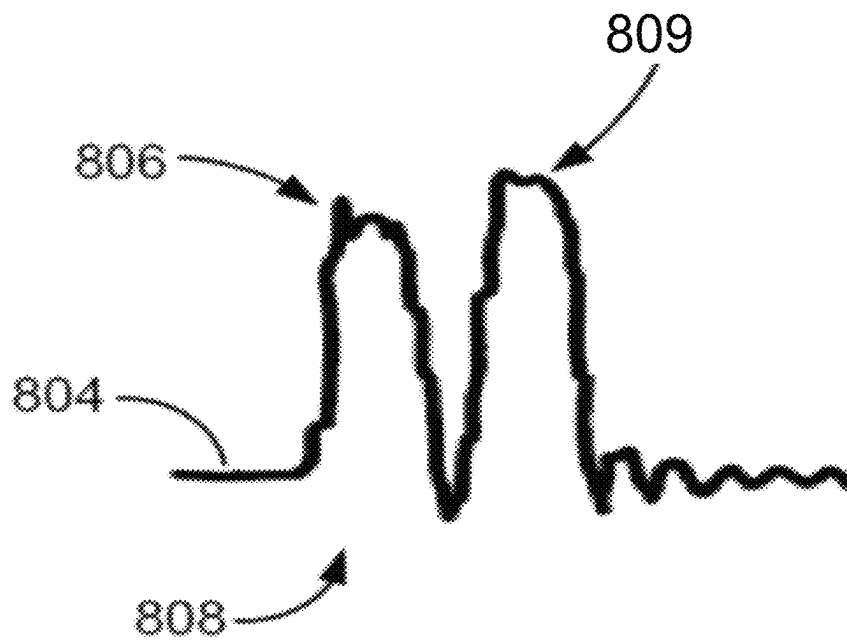


FIG. 9

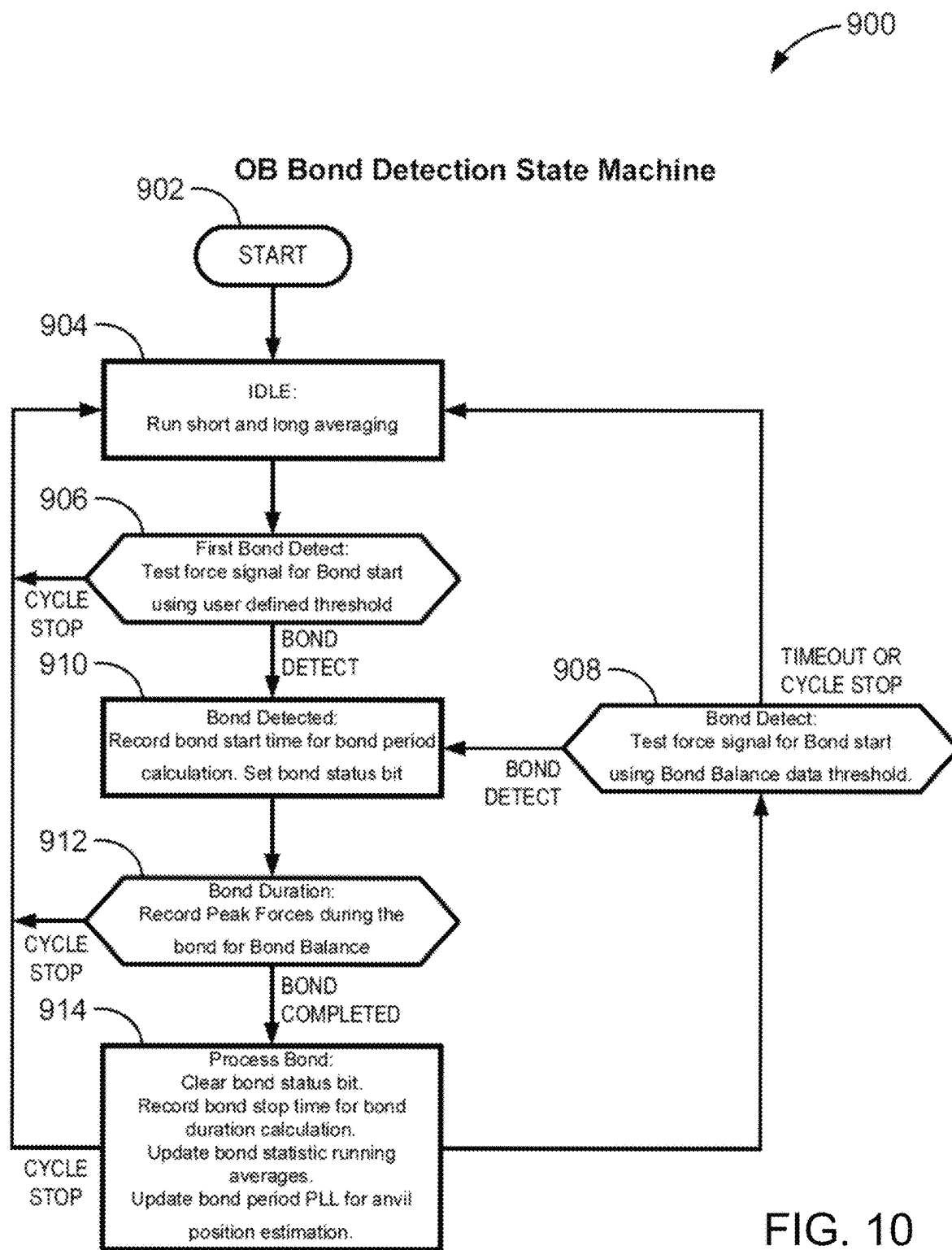


FIG. 10

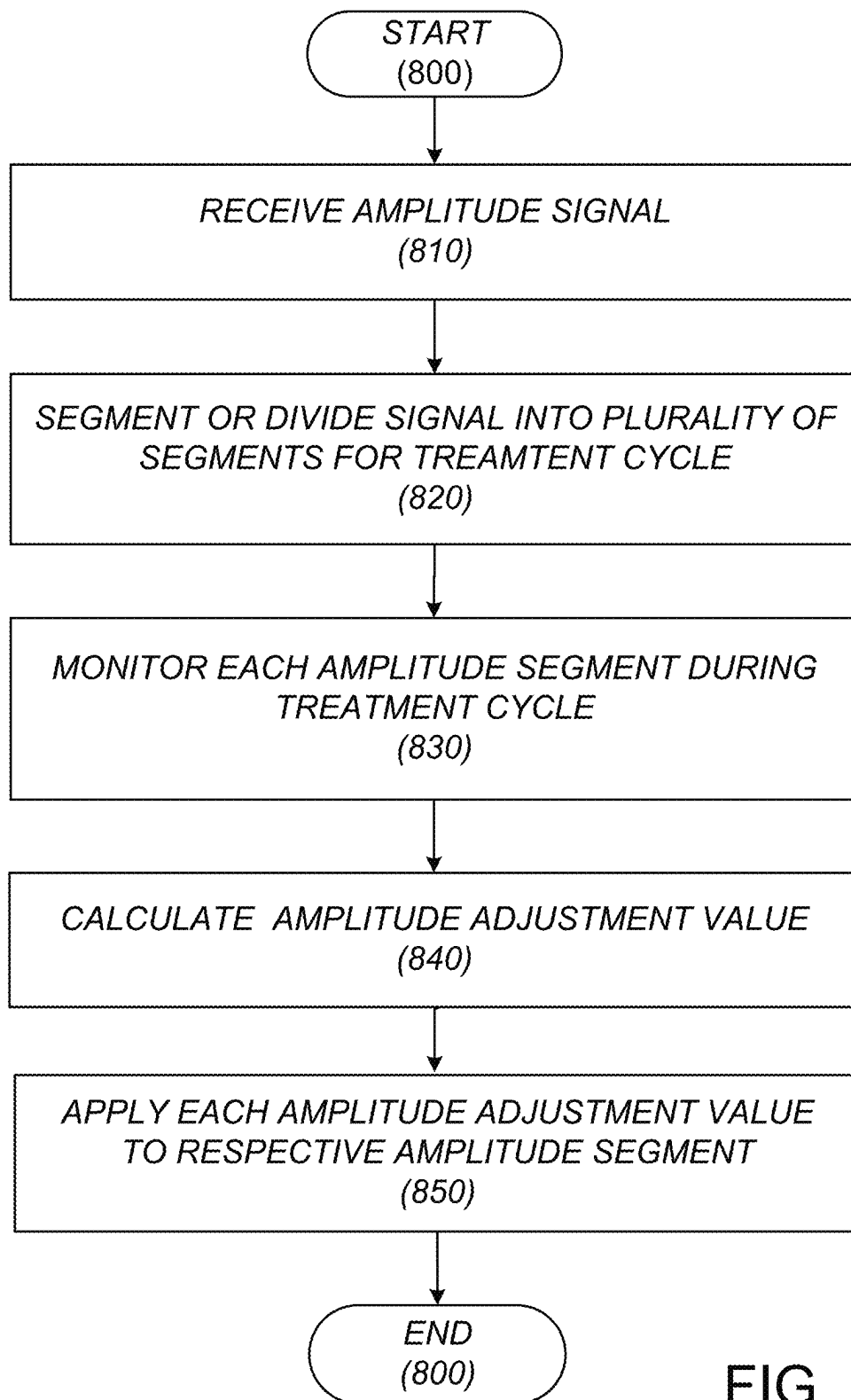
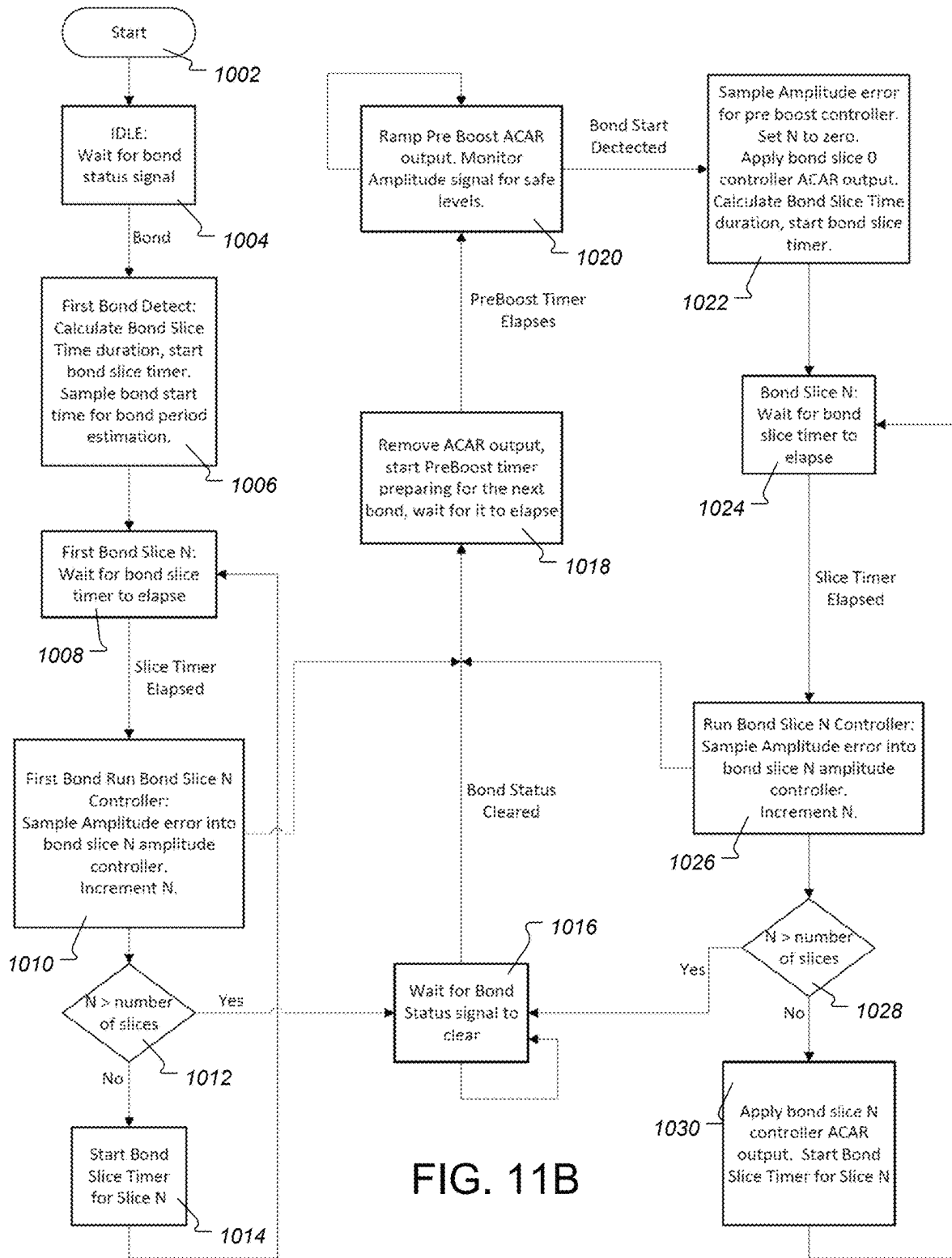
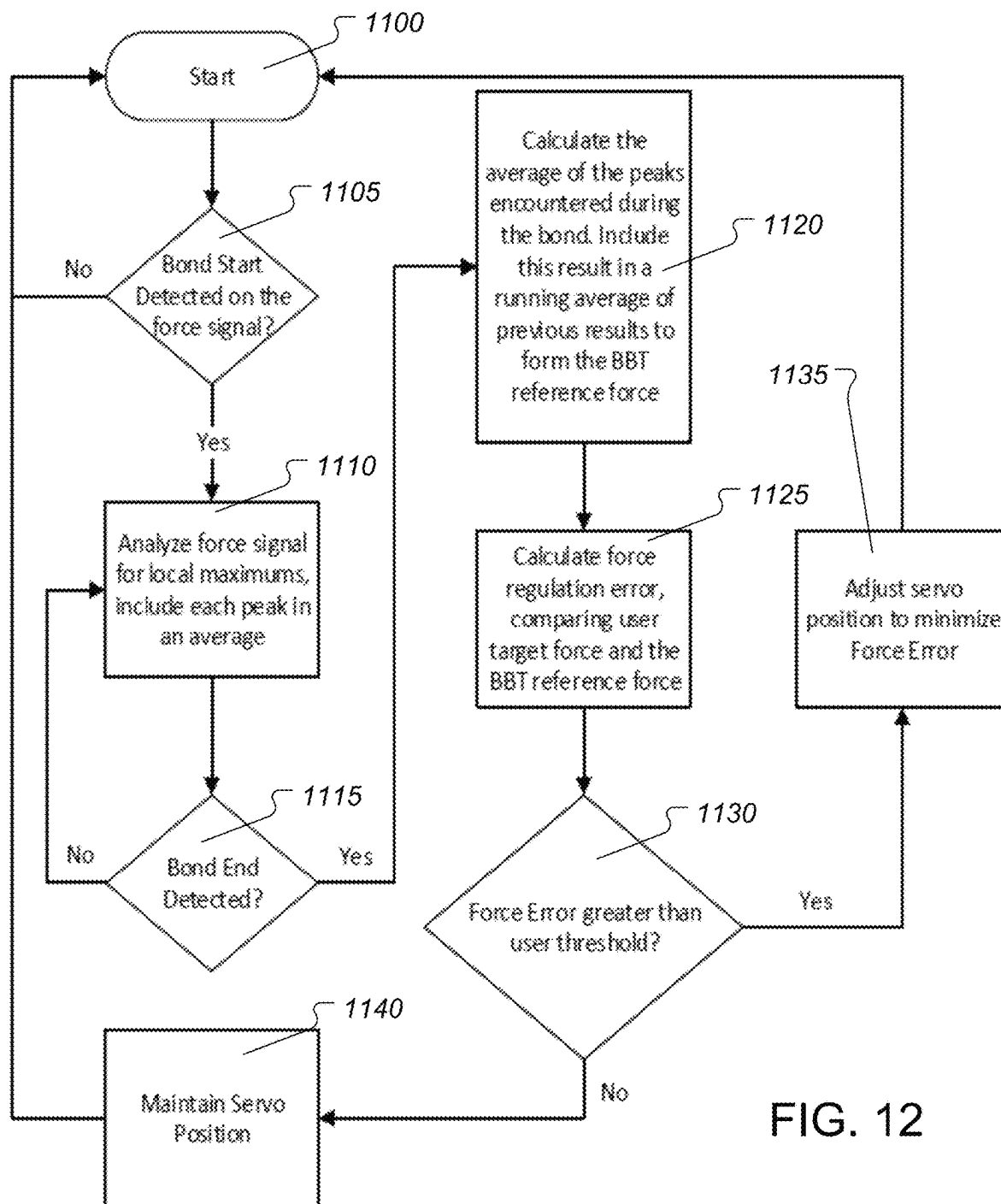
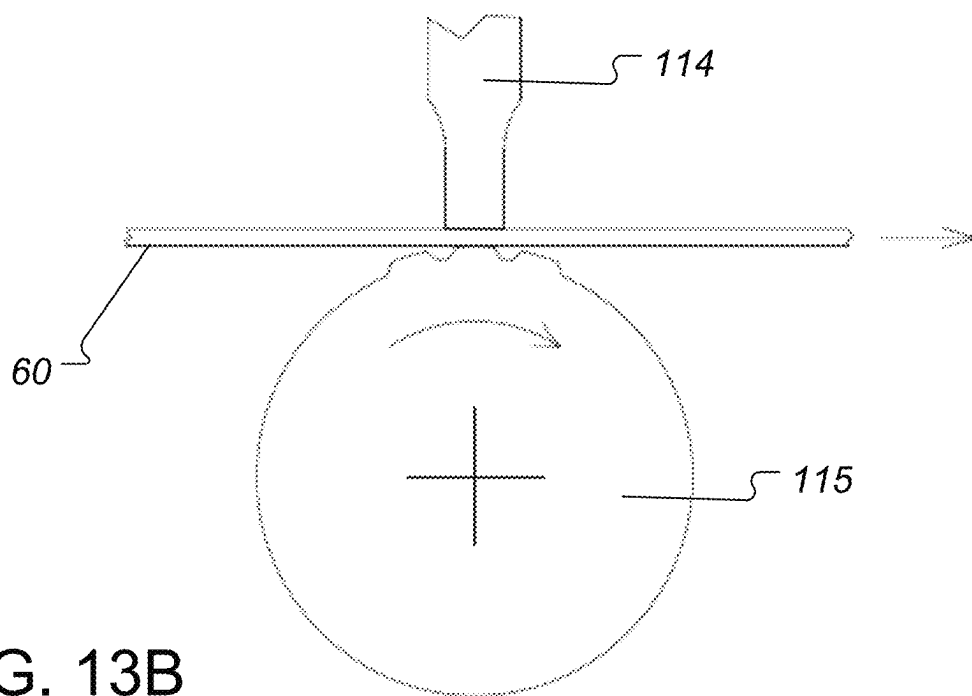
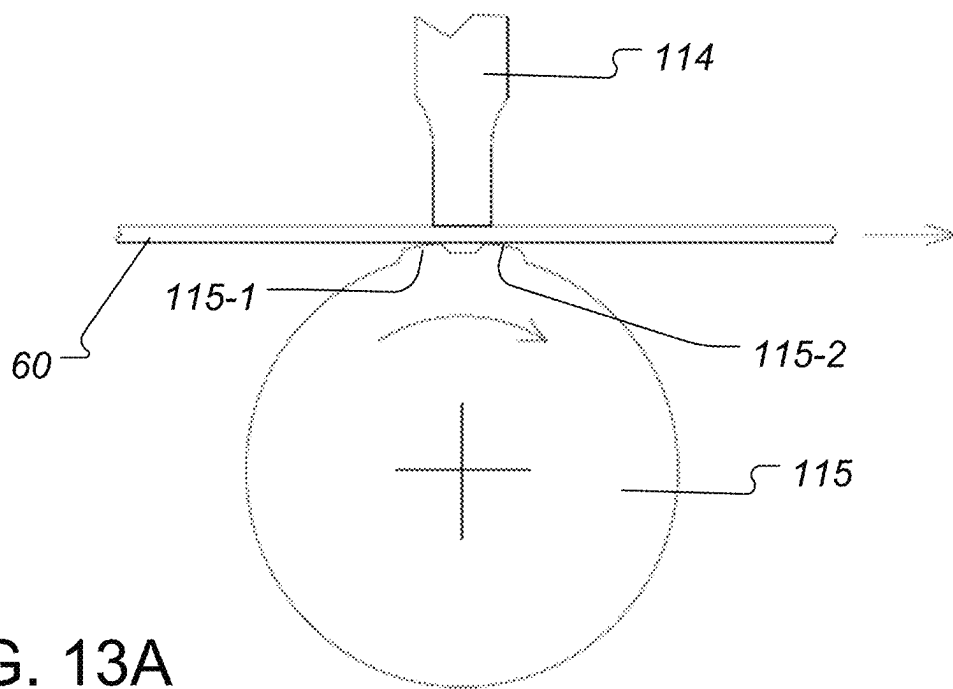


FIG. 11A







APPARATUS AND A METHOD FOR BONDING WEBS OF NON-WOVEN PLASTIC MATERIAL

CROSS REFERENCE TO PRIOR APPLICATION

This application is a bypass continuation of International Patent Application No. PCT/US2025/020281, filed on Mar. 17, 2025, which claims priority to, and the benefit thereof, provisional U.S. Patent Application Ser. No. 63/566,757, filed on Mar. 18, 2024, the entirety of both are hereby incorporated herein by reference as if fully set forth herein.

FIELD OF THE INVENTION

The present invention relates generally to nonwoven materials and, more particularly, to an apparatus, a system, and a method for treating one or more workpieces, including, but not limited to, a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation.

BACKGROUND OF THE INVENTION

State of the art technologies are available for producing ultrasonically bonded cross seals, end seals or side seams on disposable undergarments made of nonwoven materials. However, those technologies perform poorly at high production speeds. For example, at production speeds above 175 to 200 units or articles per minute, the units or articles exhibit weak, uneven or inconsistent bonds. This poor performance is attributed in part to slow sample rates and response times of existing control systems with respect to both motion and ultrasonic performance. Typical sample rates in current PLC based systems are between a few milliseconds and 20 milliseconds. Moreover, the technology exhibits lower yields, inconsistent wear of metal parts, premature failure of components, and a need for operator configuration and parameter tweaking during production.

Typically, these bonds related to cross sealing occur in pairs, happening in rapid succession between comparatively long periods of unbonded material. The percentage of bonded material per product is very low, typically <5%. The first bond in each pair is part of product X and second bond in each pair is part of product X+1, corresponding to the left side and right side of each product. At high production speeds the existing state of the art fails to distinguish the individual bonds from each other. Critical process information is lost due to slow sampling rate, treating the two bonds as a single aggregated event.

Since the poor product quality at high speeds is due, at least in part, to leading bonds in resultant articles being weaker than trailing bonds, machine operators tend to compensate by increasing the bond force control parameter, often resulting in overwelded trailing bonds. However, this strategy produces greater force and stress on the machine. Reducing or eliminating overwelding allows the machine to last longer and to extend wear and tear maintenance intervals.

An ability to produce a consistent bond force between the leading and trailing bond can overcome these and other disadvantages, resulting in consistent bonds, higher product quality, higher yield, higher throughput, lower wear and tear, and no need for operator adjustment of any configuration or parameters. The instant disclosure provides a technology that provides consistent bonds with process improvements

of six sigma quality levels, compared to state-of-the-art systems that can optimize the process to only about 3-4 sigma quality levels.

SUMMARY OF THE INVENTION

The instant disclosure provides a technology (including a system, a method, and an apparatus) for bonding nonwoven materials. The technology provides advantages such as, for example: bond strength variability of less than 10%; both sides (leading and trailing) of a bond being near equal, without any weak or uneven bonds; faster sampling rates, including taking an average of peak forces of each bond instead of sampling to peak force of multiple bonds; automated cycle rate detection without any need for external cycle rate feedback; actuator out of adjustment detection using, for example, dual encoder feedback; actuator over-travel detection; automated operator notification, including system status alerts such as maintenance required warning for mechanical actuation linkage; actual linear displacement of the horn; operability for intermittent bonding applications; and metal contact detection.

According to an aspect of the disclosure, an apparatus is provided having two or more treatment modules positionable relative to each other, wherein at least one of the treatment modules includes a joining device for treating one or more workpieces during a treatment operation. The apparatus comprises a controller arranged: to determine a force applied to at least one portion of each of the one or more workpieces during each bond in a plurality of bonds of the workpiece; analyze the measured force to determine a force characteristic value for each bond; calculate a force adjustment value based on the force characteristic value for at least one of the plurality of bonds; and apply the force adjustment value to at least one of the treatment modules. The treatment operation can include at least one of a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation. The action of applying the force adjustment value to the at least one of the treatment modules can reduce bond strength variability. In an embodiment the controller can be arranged for closed loop control.

In an embodiment the controller can be further arranged to: receive a force signal from a sensor that is indicative of the force applied to the at least one portion of each of the one or more workpieces during each bond, wherein the force is determined based on the received force signal; and adjust a position of one of the two or more treatment modules when the force adjustment value exceeds a predetermined threshold. The force characteristic value for each bond can be determined based on, for a predetermined period of time, at least one of: a peak of force values in the force signal; an average of force values in the force signal; a median of force values in the force signal; a range of force values in the force signal; a variance of force values in the force signal; and a standard deviation of force values in the force signal. The predetermined period of time can include a bond duration.

In various embodiments, the force adjustment value can be calculated based on at least one of: an average of the force characteristic values of two or more bonds; a median of the force characteristic values of two or more bonds; a range of the force characteristic values of two or more bonds; a variance of the force characteristic values of two or more bonds; and a standard deviation of the force characteristic values of two or more bonds.

In various embodiments, the controller can be arranged to: receive a force signal from a sensor, wherein the force

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characteristic value for each bond is determined based on the force signal; and determine a range of force characteristic values based on the force characteristic values for a plurality of bonds.

In various embodiments, the joining device can include an ultrasonic horn arranged to apply ultrasonic energy to the workpiece. The controller can be arranged to: receive at least one of an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal; and calculate at least one of a bond period and a bond duration based on the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal.

In various embodiments the controller can be arranged to: receive at least one of an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal; and calculate the force characteristic value for each bond based on the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal.

In an embodiment, the controller can be arranged to: receive at least one of an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal; and calculate at least one of a bond period and a bond duration based on the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal.

In an embodiment, the ultrasonic amplitude signal, the ultrasonic power signal, and/or the ultrasonic frequency signal can be received from an ultrasonic controller.

In various embodiments the controller is arranged to receive a position signal from a position sensor. The position sensor can include one or more linear encoders.

In various embodiments the range of force characteristic values can be calculated as a difference between a minimum force characteristic value and a maximum force characteristic value amongst a plurality of force characteristic values. The range of force characteristic values can correlate to bond quality amongst the plurality of bonds in the workpiece. The controller can be arranged to generate an error signal based on the range of force characteristic values. In certain embodiments the controller can be arranged to generate an alarm and/or adjust one or more parameters in at least one of the first and second treatment modules to reduce the error signal.

The workpiece can include one or more continuous webs or discrete pieces of material.

According to another aspect of the disclosure, a method is provided for treating one or more workpieces during a treatment operation comprising at least two or more treatment modules positionable relative to each other. The method can be computer-implemented. The method comprises: determining a force applied to at least one portion of each of the one or more workpieces during each bond in a plurality of bonds of the workpiece; analyzing the measured force to determine a force characteristic value for each bond; calculating a force adjustment value based on the force characteristic value for at least one of the plurality of bonds; and applying the force adjustment value to at least one of the treatment modules. The treatment operation can include at least one of a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation and applying the force adjustment value to the at least one of the treatment modules reduces bond strength variability.

According to a further aspect of the disclosure, an apparatus is provided having a welding device for treating a workpiece. The apparatus comprises: a first treatment mod-

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ule; a second treatment module positionable in proximity to the first treatment module; a toggle mechanism coupled to one of the first treatment module and the second treatment module, the toggle mechanism having an input axis; a first position sensor arranged to monitor a position of the first treatment module or the second treatment module; a second position sensor arranged to monitor a position of the toggle mechanism input axis; and a controller arranged to receive a first position signal from the first position sensor and a second position signal from the second position sensor and detect an operating condition of the first treatment module or the second treatment module based on the first position signal and the second position signal. The operating condition can include an alignment along the travel path of the first treatment module or the second treatment module with respect to an output axis. The toggle mechanism can be arranged to move said one of the first treatment module and the second treatment module with respect to the output axis. The treating can include a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation. The toggle mechanism can be arranged to adjust a travel path of said one of the first treatment module and the second treatment module to be parallel to the output axis. The toggle mechanism can include an adjustable cam.

In various embodiments the first position sensor can include one or more linear position sensors arranged to detect a position of a welding device and output the first position signal. The apparatus can include at least one backlash compensation spring arranged to bias a backlash in a direction of a working load, in parallel with the working load along the output axis.

In various embodiments the controller can be arranged to determine the travel path of the welding device based on the second position signal. The controller can be arranged to generate an alarm when the travel path exceeds a predetermined threshold value past an inversion point of the toggle mechanism. The inversion point can be detected when the direction of motion of the output axis is opposite the direction of motion of the input axis as determined by monitoring the said first position sensor and said second position sensor. The predetermined threshold value can be, for example, 600 μm or greater along the input axis.

In various embodiments the apparatus can include a force sensor arranged to measure a force applied to one or more links in the toggle mechanism and output a force signal comprising a measured force value. The controller can be arranged to receive the force signal from the force sensor and calculate an external load by subtracting a force exerted by the at least one backlash compensation spring from the measured force value. The first treatment module can include an anvil and the second treatment module can include the welding device. The anvil can be a rotary anvil.

In various embodiments the welding device comprises an ultrasonic horn. The controller can be arranged to perform a Bond Balance Technology (BBT) operation in an intermittent bonding application or a continuous bonding application. The BBT operation can include recognizing a plurality of individual bond force peaks, recording a value for each bond force peak, and averaging two or more of the values to determine a force regulation value and/or a position regulation reference value.

The operating condition can include metal-to-metal contact and the controller can be arranged to perform metal-to-metal contact detection operation between an anvil and the welding device.

In various embodiments the apparatus can include an alarm indicator that emits a visible or audible alarm. The

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controller can be arranged to generate an alarm signal based on the operating condition and send the alarm signal to the alarm indicator. The controller can be further arranged to hold an actuator at a linear position that is within 100 μm , or less, of metal-to-metal contact. The alarm indicator can include a human-machine interface (HMI).

In various embodiments the controller can be arranged with Adaptive Cyclic Amplitude Regulation (ACAR) to provide ultrasonic amplitude regulation and control power delivery to the workpiece by determining a real-time amplitude adjustment based on historical amplitude response to reduce amplitude deviations prior to, during, and after bonding.

In various embodiments the controller can be arranged to control the welding device to create a space for a splice in the workpiece.

According to further aspect of the disclosure, an apparatus is provided for bonding a workpiece. The apparatus comprises: a welding device arranged to apply energy to the workpiece; an anvil having an anvil pattern that includes a plurality of projections; at least one position sensor arranged to measure at least one position of the welding device and output at least one position signal; a force sensor arranged to measure a force applied to the welding device and output a force signal representative of the applied force; and a controller arranged to receive the position signal and the force signal and detect an operating condition of the welding device. A position of at least one of the welding device and the anvil can be adjusted based on the operating condition to minimize bond strength variability for intermittent projections on the anvil pattern. The controller can be arranged to sample the force signal and, based on the sampled force signal, calculate a force regulation error. A position of the welding device can be adjusted when the force error exceeds a predetermined threshold. The operating condition can include at least one of a peak force, an average force, a bond period, a bond duration, and a power level. The position sensor can include one or more linear encoders.

In an embodiment the apparatus can include a toggle mechanism having a cam, wherein the controller is arranged to detect movement of the cam.

In various embodiments the controller can be arranged to perform the Bond Balance Technology (BBT) operation to track to an average of multiple bond peak forces. The controller can be arranged to control a bond force toward a selectable force. The controller can be arranged: to control an overall bond peak force or an average of peak forces of individual bonds for intermittent applications; to maintain a constant or variable force for continuous applications; to detect a treating cycle rate based on a timing analysis of the force signal, wherein the treating cycle rate comprises a rate (units/minute) at which each workpiece in a plurality of workpieces is treated, wherein the treatment comprises a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation; and/or with Adaptive Cyclic Amplitude Regulation (ACAR) to provide ultrasonic amplitude regulation and control power delivery by the welding device (114) to the workpiece (60).

In various embodiments ACAR can reduce amplitude deviations during bonds by matching ultrasonic regulation control response to the individual load characteristics of the distinct bonds occurring in rapid succession.

In an embodiment the apparatus can include another position sensor arranged to measure alignment of a travel path of the welding device and output another position signal. The operating condition can include the movement of the cam. The controller can be arranged to: monitor the

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position signal and the another position signal and determine whether the movement of the cam is due to an adjusting force applied by an operator; and/or monitor the position signal and the another position signal and determine whether the movement of the cam is due to a force other than the adjusting force applied by the operator. The operating condition can include metal-to-metal contact and the controller can be arranged to perform metal-to-metal contact detection operation between the anvil and the welding device.

In various embodiments the apparatus includes an alarm indicator that emits a visible or audible alarm, wherein the controller is arranged to generate an alarm signal based on the operating condition and send the alarm signal to the alarm indicator, and hold the actuator at a linear position that is within 100 μm , or less, of metal-to-metal contact. The metal-to-metal contact can be detected based on detecting an increase in standard deviation of an ultrasonic output frequency of an ultrasonic energy generator, such increase being at least 10%. The metal-to-metal contact can be detected based on detecting an increase in standard deviation of an ultrasonic output frequency lock quality of the ultrasonic energy generator, such increase being at least 10%. At least one of the position sensor and the another position sensor can include a motor encoder or linear encoder. The controller can be arranged to monitor a backlash distance based on the position signal and said another position signal. The controller can be arranged to monitor a backlash distance based on the position signal from a first position sensor and said another position signal from a second position sensor, wherein the backlash distance is monitored during the application of intermittent force.

According to another aspect of the disclosure, an apparatus is provided having a first treatment module and a second treatment module positionable in proximity to the first treatment module, wherein at least one of the first treatment module and the second treatment module includes a welding device for treating one or more workpieces, the apparatus comprising a controller arranged to: measure an edge force at each of a plurality of first edges of a workpiece and an edge force at each of a plurality of second edges of an adjoining workpiece; analyze the first plurality of edge forces and the second plurality of edge forces to determine a peak trailing edge force and a peak leading edge force; determine a force adjustment value based on the peak trailing edge force and the peak leading edge force; and apply the adjustment value to at least one of the first treatment module and the second treatment module. At least one of the plurality of first edges or at least one of the plurality of second edges can include a trailing edge or a leading edge, wherein the treating can comprise a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation. The force adjustment value can be determined based on an average or a mean value of the peak trailing edge force and the peak leading edge force.

According to yet another aspect of the disclosure an apparatus is provided having a first treatment module and a second treatment module positionable in proximity to the first treatment module, wherein at least one of the first treatment module and the second treatment module includes a welding device for treating one or more workpieces, the apparatus comprising a controller arranged to: receive one or more signals, including at least one of a force signal, an ultrasonic amplitude signal, an ultrasonic power signal, an ultrasonic frequency signal, and a position signal; calculate a plurality of bond force peak values based on the received one or more signals; and calculate a bond force span value

based on the plurality of bond force peak values. The bond force span value is calculated as a difference between a minimum bond force peak value and a maximum bond force peak value during a treatment operation. The bond force value is monitored during the treatment operation as a bond balance error signal for process consistency. The bond force span value can include a range of force values during a treatment operation.

According to yet another aspect of the disclosure an apparatus is provided having a first treatment module and a second treatment module positionable in proximity to the first treatment module, wherein at least one of the first treatment module and the second treatment module includes a welding device for treating one or more workpieces, the apparatus comprising a controller arranged to: receive a real-time amplitude signal of ultrasonic vibration of at least one of the first treatment module and the second treatment module during a treatment cycle; segment the amplitude signal into a plurality of amplitude segments for the treatment cycle; monitor an amplitude value of each of the plurality of amplitude segments during the treatment cycle; operate a plurality of closed-loop control algorithms to determine a plurality of amplitude adjustment values, each of the plurality of amplitude adjustment values corresponding to a respective one of the plurality of amplitude segments; and apply each amplitude adjustment value to the corresponding respective one of the plurality of amplitude segments during the subsequent treatment cycle in real-time. The treatment cycle can include treating a single workpiece in a plurality of workpieces. The treating can comprise a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation. The controller can be arranged to: generate a proportional-integral-derivative (PID) setting for each of the plurality of amplitude segments; and apply each PID setting to the respective one of the plurality of amplitude segments.

The apparatus can further comprise: a force sensor arranged to measure a force applied to at least one of the first treatment module and the second treatment module and output a force signal representative of the applied force; an amplitude sensor arranged to detect and measure ultrasonic vibration in real-time and output the ultrasonic amplitude signal; a power sensor arranged to detect and measure real-time power supplied to, or applied by, at least one of the first treatment module and the second treatment module; and at least one position sensor arranged to detect and measure a real-time position of at least one of the first treatment module and the second treatment module, including a real-time position of the welding device. The controller can be arranged: to detect a treating cycle rate based on a timing analysis of at least one of the force signal, the ultrasonic amplitude signal, the ultrasonic power signal, and the at least one position signal, wherein the treating cycle rate comprises a rate (units/minute) at which each workpiece in a plurality of workpieces is treated; and/or with Adaptive Cyclic Amplitude Regulation (ACAR) to provide ultrasonic amplitude regulation and control power delivery to the workpiece by determining a real-time amplitude adjustment based on historical amplitude response to reduce amplitude deviations prior to or during bonding. Each of the plurality of closed-loop control algorithms can include a separate Proportional-Integral-Derivative (PID) tuning loop corresponding to a respective one of the plurality of amplitude segments.

In various embodiments the controller can be arranged to: receive one or more signals, including at least one of the force signal, the ultrasonic amplitude signal, the ultrasonic

power signal, and the at least one position signal; calculate a plurality of bond force peak values based on the received one or more signals; and calculate a bond force span value based on the plurality of bond force peak values. The bond force span value can be calculated as a difference between a minimum bond force peak value and a maximum bond force peak value during a treatment operation. The bond force value can be monitored during the treatment operation as a bond balance error signal for process consistency.

According to a still further aspect of the disclosure an apparatus is provided having at least a first treatment module and a second treatment module positionable in proximity to the first treatment module, wherein at least one of the first treatment module and the second treatment module includes a welding device for treating one or more workpieces. The apparatus comprises: a force sensor arranged to measure a force applied to at least one of the first treatment module and the second treatment module and output a force signal representative of the applied force; an amplitude sensor arranged to detect and measure the real-time ultrasonic vibration and output the ultrasonic amplitude signal representative of the real-time ultrasonic vibration; a power sensor arranged to detect and measure real-time power supplied to, or applied by, at least one of the first treatment module and the second treatment module and output the ultrasonic power signal representative of the real-time power; and at least one position sensor arranged to detect and measure a real-time position of at least one of the first treatment module and the second treatment module, including a real-time position of the welding device and output at least one position signal. A treating cycle rate can be detected based on a timing analysis of at least one of the force signal, the amplitude signal, the power signal, and the at least one position signal. The treating cycle rate can include a rate (units/minute) at which each workpiece in a plurality of workpieces is treated. The apparatus can comprise a controller. The controller can be arranged to: receive one or more signals, including at least one of the force signal, the ultrasonic amplitude signal, the ultrasonic power signal, and the at least one position signal; calculate a plurality of bond force peak values based on the received one or more signals; and calculate a bond force span value based on the plurality of bond force peak values. The bond force span value can be calculated as a difference between a minimum bond force peak value and a maximum bond force peak value amongst a plurality of bond force peak values during a treatment operation. The bond force value can be monitored during the treatment operation as a bond balance error signal for process consistency.

Additional features, advantages, and embodiments of the disclosure may be set forth or apparent from consideration of the detailed description and drawings. Moreover, it is to be understood that the foregoing summary of the disclosure and the following detailed description and drawings provide nonlimiting examples that are intended to provide further explanation without limiting the scope of the disclosure as claimed.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying drawings, which are included to provide a further understanding of the invention, are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and together with the detailed description serve to explain the principles of the invention.

FIG. 1A shows a block diagram of an embodiment of a workstation constructed according to the principles of the disclosure.

FIG. 1B shows an embodiment of a controller constructed according to the principles of the disclosure, which can be included in the workstation in FIG. 1A.

FIG. 2 shows an embodiment of the welding system constructed according to the principles of the disclosure, which can include the workstation in FIG. 1.

FIG. 3 shows a portion of the welding system in FIG. 2 with certain components removed to show certain covered features in the welding system.

FIG. 4 shows the welding system in FIG. 2 with the frame removed.

FIG. 5 shows an embodiment of a thruster that can be included in the welding system of FIG. 1.

FIG. 6 shows calibration components that can be included in the welding system of FIG. 1.

FIGS. 7A and 7B show two different tensile strength plots from trailing to leading edges, including without Bond Balance Technology (BBT) in FIG. 7A and with the BBT in FIG. 7B.

FIG. 8A shows plot diagrams of instantaneous ultrasonic amplitude during an intermittent welding application with a bond duration of 15 mS.

FIG. 8B shows ultrasonic amplitude plot diagrams in which Adaptive Cyclic Amplitude Regulation (ACAR) has been enabled.

FIG. 9 shows a typical force signal of a dual bond intermittent welding application.

FIG. 10 shows a bond detection process, according to the principles of the disclosure.

FIGS. 11A and 11B show nonlimiting embodiments of an ACAR process, according to the principles of the disclosure.

FIG. 12 shows a Bond Balance Technology (BBT) process, according to the principles of the disclosure.

FIGS. 13A and 13B show two nonlimiting embodiments of a joining device and an anvil positioned for treating a workpiece moving in a direction depicted by an arrow.

The present disclosure is further described in the detailed description that follows.

DETAILED DESCRIPTION OF THE INVENTION

The invention and its various features and advantageous details are explained more fully with reference to the non-limiting embodiments and examples that are described or illustrated in the accompanying drawings and detailed in the following description. It is noted that features illustrated in the drawings are not necessarily drawn to scale, and features of one embodiment can be employed with other embodiments, as those skilled in the art will recognize, even if not explicitly stated. Descriptions of well-known components and processing techniques may have been omitted so as to not unnecessarily obscure the embodiments of the invention. The examples are intended merely to facilitate an understanding of ways in which the invention can be practiced, and to further enable those skilled in the art to practice the embodiments of the invention. Accordingly, the examples and embodiments should not be construed as limiting the scope of the invention. Moreover, it is noted that like reference numerals represent similar parts throughout the several views of the drawings.

FIG. 1A shows an embodiment of a workstation 1 constructed according to the principles of the disclosure. In various embodiments, the workstation 1 can be configured

as a single machine or device, or a plurality of machines or devices, comprising components or modules configured to interact and operate as described below. The workstation 1 includes at least one of a first treatment module 10 and a second treatment module 20, a driver module 30, a sensor module 40 and a controller 50. In various embodiments the controller 50 can be configured as seen in FIG. 1B. In certain embodiments the driver module 30 can be incorporated in the controller 50, for example, as the driver suite 250 (shown in FIG. 1B).

In certain embodiments the controller 50 can communicate with one or more controllers (not shown), which can be provided external to the workstation 1, or provided in at least one of the first and second treatment modules 10, 20, and configured to receive at least one of a force signal, an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal from the one or more controllers (not shown). The workstation 1 can be configured to receive one or more workpieces 60 for treatment, such as, for example, bonding, welding, soldering, fusing, or cutting. For instance, two or more workpieces 60 can be bonded, welded, soldered, or fused together by the first treatment module 10 and/or the second treatment module 20.

In various applications, the workpiece(s) 60 can include, for example, one or more continuous webs or discrete pieces of material. The workpiece(s) 60 can include, for example, two or more layers of nonwoven material by treatment. The treatment can include, for example, bonding, welding, or fusing the two or more layers of nonwoven material together. The treatment can further include cutting the bonded, welded, or fused layers into one or more products having a predetermined shape and dimensions (for example, width, length, thickness).

In other applications, the workpiece(s) 60 can include one or more materials or products to be treated. The treatment can include, for example, bonding, welding, or fusing the workpiece 60 to a material, an object, or another workpiece 60. The treatment can include cutting the workpiece(s) 60 to a predetermined shape or dimensions, including, for example, width, height, length.

In various embodiments, either or both the first treatment module 10 and the second treatment module 20 each can include one or more joining devices such as, for example, a sonotrode (or horn), a cylindrical sonotrode (or horn), a blade sonotrode (or horn), a block sonotrode (or horn), a rotary sonotrode (or horn), a laser welder, an arc welder, a resistance metal inert gas (MIG) welder, a plasma arc welding (PAW) welder, a tungsten inert gas (TIG) welder, an anvil, a rotary anvil, a fixed shaped nest, or other device capable of treating one or more workpieces 60, and/or holding the one or more workpieces 60 during treatment. The joining device(s) can have a shape and dimensions (including, for example, width, length height) contoured to the workpiece(s) 60 to be treated.

One of the first treatment module 10 and the second treatment module 20 can include a joining device (for example, a sonotrode) or a cutting device capable of transferring energy (for example, ultrasonic energy) to the workpiece 60 being treated, including welding, soldering, fusing, or cutting; and the other of the first treatment module 10 and the second treatment module 20 can include an anvil, such as, for example, an anvil 115 (shown in FIG. 13A or 13B). The first module 10 and the second module 20 can be provided opposite each other and aligned with the workpiece 60 to be treated.

At least one of the first treatment module 10 and the second treatment module 20 can include one or more sensors

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(for example, position sensor, voltage sensor, current sensor, frequency sensor, temperature sensor, pressure sensor, humidity sensor, or the like), which can be incorporated into as a sensor module (not shown).

At least one of the first treatment module **10** and the second treatment module **20** can include one or more drivers (for example, a DC motor driver, an AC motor driver, a variable-frequency driver, a servo-motor driver, a stepper motor driver, a pneumatic driver, or the like) configured to drive one or more components in the module(s) **10**, **20**. The components can include, for example, a DC motor, an AC motor, a variable-frequency motor, a servomotor, a stepper motor, a pneumatic actuator, or the like.

In various embodiments, the first treatment module **10** and/or the second treatment module **20** can include a rotary joining device (not shown) configured to rotate about an axis or a fixed joining device (not shown) configured to move in a single plane along a linear axis. The first treatment module **10** and/or the second treatment module **20** can include a combination of one or more rotary joining devices and/or fixed joining devices. The joining device can include one or more ultrasonic horns and/or one or more anvils (for example, anvil **115** shown in FIG. **13A** or **13B**). The one or more ultrasonic horns can be configured to perform a treatment operation such as, for example, joining, welding, or cutting one or more materials.

In various embodiments, the first treatment module **10** includes one or more anvils (for example, **115** shown in FIG. **13A** or **13B**) and the second treatment module **20** includes one or more joining devices. The joining device(s) can include, for example, an ultrasonic horn. The joining device(s) can be configured for joining, welding, adhering, or cutting material.

The driver module **30** can include an energy source configured to supply energy and/or material to the first treatment module **10** and/or the second treatment module **20** to power, operate, or control the joining device(s), including, for example, any combination of electricity, gas, fluid, or material.

In various embodiments, the driver module **30** can include any one or more of, for example, a motor, a linear servo actuator, a servomotor, a rotary servo motor, a stepper motor coupled to a rotary-to-linear converter, a servomechanism, a linear motor, a stepper motor, a minor loop feedback motor, a pneumatic actuator, a hydraulic actuator, or other drive mechanism configured to transform energy to a linear, rotary, or oscillatory motion, including one or more mechanisms that can be operated or controlled by means of an electric current or voltage, a pressurized fluid (for example, a gas or a liquid).

In certain embodiments, the driver module **30** includes a pair of driver modules, one of which is configured to drive the first treatment module **10** and the other of which is configured to drive the second treatment module **20**. In an embodiment one of the pair of driver modules can be configured to rotate the joining device in the first treatment module **10**. The other of the pair of driver modules can be configured to rotate the joining device in the second treatment module **20**.

In an embodiment, the driver module **30** includes an actuator **102** and/or a ram module **116** (shown in FIG. **2**).

The sensor module **40** can include one or more sensors configured to detect, measure, or monitor components or properties in various parts of the workstation **1**. The sensor module **40** can include one or more position sensors con-

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figured to detect, measure, and monitor a position of the first treatment module **10**, the second treatment module **20**, or one or more workpieces **60**.

In an embodiment, the sensor module **40** includes a linear position sensor **204** (shown in FIG. **3**), an actuator position sensor **312** (shown in FIG. **4**), and a force sensor **306** (shown in FIG. **4**).

The controller **50** can include one or more processors configured for operation and control of the workstation **1**, including one or more of the first treatment module **10**, the second treatment module **20**, the driver module **30**, the sensor module **40**, and the workpiece(s) **60**. In various embodiments, the controller **50** can include a Programmable Logic Controller (PLC) (not shown). The PLC (not shown) can include a power supply, an input module, a processor, a programming device, and an output module. The one or more processors are configured to control the interactions of the various components, including the interaction between the first treatment module **10**, the second treatment module **20**, the driver module **30**, and the workpiece(s) **60**, such that the one or more workpieces **60** are supplied to and positioned in a predetermined location, and held in alignment according to a predetermined configuration, proximate to the first treatment module **10** and/or the second treatment module **20**; then the first treatment module **10** and/or the second treatment module **20** are operated and controlled to treat the workpiece(s) **60** according to a predetermined treatment, such as, for example, bonding, welding, soldering, fusing, or cutting.

The Controller

FIG. **1B** illustrates a block diagram of an embodiment of a controller **200** constructed according to the principles of the disclosure. In various embodiments the controller **50** (shown in FIG. **1A**) can be configured as the controller **200**.

In certain embodiments the controller **200** can be included in at least one of the first treatment module **10** and the second treatment module **20**, or provided external to the workstation **1** (shown in FIG. **1A**) and connected through one or more wired or wireless connections. The controller **200** includes a bus **205**, a processor **210**, a memory **220**, an input-output (IO) interface **230**, a communication unit **240**, and a driver suite **250**. Any of the components **210** to **250** can be interconnected using various buses, including the bus **205**, and can be mounted on a common motherboard or in another manner, as appropriate.

The processor **210** can be arranged to process instructions for execution within the controller **200**, including instructions stored in the memory **220**. The processor **210** can include any of various commercially available processors. Dual microprocessors and other multi-processor architectures can be employed as the processor **210**. The processor **210** can include a central processing unit (CPU), an application-specific integrated circuit (ASIC), or a graphic processing unit (GPU). The processor **210** is arranged to interact with all of the components in the controller **200** to carry out or facilitate the processes described herein.

The bus **205** can include any of several types of bus structures that can further interconnect to a memory bus (with or without a memory controller), a peripheral bus, and a local bus using any of a variety of commercially available bus architectures.

The memory **220** includes a read-only memory (ROM) **220A** and a random-access memory (RAM) **220B**. In certain embodiments, the memory **220** can include a hard disk drive (HDD) **220C**. The memory **220** can provide nonvolatile storage of data, data structures, and computer-executable instructions, and can accommodate the storage of any data

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in a suitable digital format. The memory **220** can include a computer-readable medium that can hold executable or interpretable computer code (or instructions) that, when executed by the processor **210**, cause the steps, processes and methods in this disclosure to be carried out.

The computer-readable medium can be contained in the memory **220**, and can include sections of computer code that, when executed by the processor **210**, cause the controller **200** to perform the processes and operations provided in this disclosure.

In an embodiment, a basic input-output system (BIOS) can be stored in the ROM **220A**, which can include a non-volatile memory, an erasable programmable read-only memory (EPROM), or an electrically erasable programmable read-only memory (EEPROM). The BIOS can contain the basic routines that help to transfer instructions and data between any one or more of the components in the controller **200**, such as during start-up.

The RAM **220B** can include dynamic random-access memory (DRAM), a synchronous dynamic random-access memory (SDRAM), a static random-access memory (SRAM), a nonvolatile random-access memory (NVRAM), or another high-speed RAM for caching data.

In certain embodiments the HDD **220C** can include, for example, an enhanced integrated drive electronics (EIDE) drive, a serial advanced technology attachments (SATA) drive, or any suitable hard disk drive. The HDD **220C** can be configured for external use in a suitable chassis (not shown).

A computer program product can be tangibly embodied in a non-transitory computer-readable medium, which can be contained in the memory **220**. The computer program product can contain instructions that, when executed by the processor **210**, cause the processor **210** to perform one or more methods, instructions, or operations, such as those included in this disclosure.

The input-output (IO) interface **230** can receive commands or data from an operator via a user interface (not shown), such as, for example, a keyboard (not shown), a touch-display (not shown), a mouse (not shown), a pointer (not shown), a stylus (not shown), an interactive voice response (IVR) system (not shown), a microphone (not shown), a speaker (not shown), or a display device (not shown). The received commands and data can be forwarded from the IO interface **230** as instruction signals and/or data signals, via the bus **205**, to any of the components in the controller **200**.

The communication unit **240** can be configured to connect to and/or exchange communication signals with external communication devices (not shown), such as, for example, display devices, speaker devices, alarm devices, computers, mobile computers, computer tablets, laptops, smartphones, force sensors, position sensors, temperature sensors, voltage sensors, current sensors, power sensors, frequency sensors, pressure sensors, or other electronic sensor devices. The communication unit **240** can include a modem, a transmitter, a receiver or a transceiver. The communication unit **240** can include a wired or a wireless communication network interface. The communication unit **240** can generate an alarm signal and output the signal to a display device or an alarm device that produces a visible and/or audible alarm.

The driver suite **250** includes one or more joining device drivers **250A**, each of which is configured to generate and send a drive signal to a corresponding joining device (for example, an ultrasonic horn) to operate the joining device(s), for example, according to the various embodiments described in this disclosure. The driver suite **250** can option-

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ally include an anvil driver **250B**, an alignment driver **250C**, and/or a workpiece driver **250D**.

The joining device driver **250A** can be configured to interact with the processor **210** and generate one or more joining device drive signals to control operation of one or more joining devices in the treatment modules **10** and/or **20** (shown in FIG. 1A), or the joining device **114** (shown in FIGS. 2-6). Referring to the embodiments depicted in FIGS. 2-6, the joining device drive signal(s) can be supplied to a ram **112** to power and control operation of the joining device **114** (for example, an ultrasonic horn), as well as the components that cause the device **114** to move along an output axis **314** (shown in FIG. 4), such as, for example, a motor (not shown). Based on the joining device drive signal(s), the processor **210** can control operation of the device **114**, including an operating frequency of the device **114**, an operating power level of the device **114**, a duration (or time) of operation of the device **114**, the travel distance of the device **114** along the output axis **314** (shown in FIG. 4), and a force applied by the device **114** to the workpiece **60** (shown in FIG. 1), including the duration of the applied force.

The optional anvil driver **250B** can be included in the embodiments of the workstation 1 comprising at least one adjustable anvil (not shown). The anvil driver **250B** can be configured to interact with the processor **210** and generate one or more anvil drive signals to control operation of a corresponding one of the one or more anvils (for example, anvil **115** shown in FIG. 13A or 13B), such as, for example, positioning and alignment of the anvil.

The optional alignment driver **250C** can be included in embodiments of a welding device **100** equipped with an automated adjustable cam **104** (shown in FIG. 4). The alignment driver **250B** can be configured to interact with the processor **210** and generate a cam drive signal to control operation of an adjustable cam **104** to move a toggle mechanism **120** (shown in FIG. 4) into alignment with the output axis **314** (shown in FIG. 4).

The optional workpiece driver **250D** can be included in the embodiments of the welding system **100** comprising an automated placement mechanism (not shown), such as, for example, feeding and/or aligning and positioning the workpiece **60** (shown in FIG. 1) for treatment by the device **114**.

Referring to FIG. 1A, the first treatment module **10** and/or the second treatment module **20** can include any of the welding devices and/or anvils described in commonly assigned U.S. Pat. Nos. 10,259,165 B2, 10,746,703 B1, 10,807,314 B1, 11,407,182 B1, and 11,254,066 B2, all of which are incorporated herein in their entireties.

50 Welding System

FIGS. 2-6 show various views of an embodiment of a welding system **100** constructed according to the principles of the disclosure. In various embodiments, the welding system **100** can include the workstation 1 (shown in FIG. 1) in which: the first treatment module **10** includes a joining device (not shown) such as an anvil or a rotary anvil (for example, anvil **115** shown in FIG. 13A or 13B); the second treatment module **20** includes a joining device **114** such as an ultrasonic horn; and the sensor module **40** includes the linear position sensor **204** (shown in FIG. 3), the actuator position sensor **312** (shown in FIG. 4), and the force sensor **306** (shown in FIG. 4).

In various embodiments the welding system **100** can include a controller **50/200** configured to reduce bond variability for intermittent projections on a treatment module **10** having anvil pattern rolls. The controller **50/200** can be configured to receive force signals or force signal samples

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from the force sensor **306**, apply a force sampling algorithm, and track to an average of multiple bond force peaks in a Bond Balance Technology (BBT), as discussed in detail below. The controller **50/200** can include, for example, a sample-and-hold circuit (not shown) that samples the force signals from the force sensor **306**. The controller **50/200** can be configured to reduce amplitude spikes, both positive and negative, during bonds (for example, as seen in FIG. 9).

In various embodiments, the controller **50/200** can be configured to monitor and detect unwanted cam movement. In certain embodiments, the controller **50/200** can monitor position signals from both position sensors **204** (shown in FIG. 3) and **312** (shown in FIG. 4) and detect if the cam moved from unwanted instantaneous high force or operator adjustment. If unwanted movement is detected, the controller **50/200** can generate a fault signal and stop machine operation.

In certain embodiments in which the treatment module **10** includes an ultrasonic horn and the treatment module **20** includes an anvil, the controller **50/200** can be configured to monitor and detect when the horn and anvil surfaces are not parallel. In this regard, the controller **50/200** can be configured to monitor position signals from both position sensors **204** and **312** to detect if the two surfaces are no longer parallel from unwanted instantaneous high forces. If the controller **50/200** determines the surfaces are not parallel based on the position signals, the controller **50/200** can generate a fault signal and stop machine operation. Another algorithm for detecting non-parallel condition between a horn and an anvil could utilize ultrasonic Power monitoring. When horn and anvil are parallel to each other the contact area is the greatest, and, at a given force, will result in the highest ultrasonic Power draw. Should these two surfaces become nonparallel at the same force, the ultrasonic Power value would decrease. Either rolling average or Standard Deviation of the Power could be used to determine the change is parallelism.

In various embodiments, the controller **50/200** can be configured for closed loop operation for bonding continuous webs with improved bond quality variability, with minimized maintenance intervals and automatic maintenance monitoring.

In the embodiment depicted in FIGS. 2-6, the welding system **100** includes a thruster assembly TA, which includes an actuator **102**, a frame **106**, the ram module **116**, the toggle mechanism **120** (shown in FIG. 4), and the linear position sensor **204** (shown in FIG. 3). The welding system **100** can include a cam **104**. The actuator **102** can include, for example, a motor, a linear servo actuator, a servomotor, a rotary servo motor, a stepper motor coupled to a rotary-to-linear converter, a servomechanism, a linear motor, a stepper motor, a minor loop feedback motor, a pneumatic actuator, a hydraulic actuator, or other drive mechanism configured to transform energy to a linear, rotary, or oscillatory motion, including one or more mechanisms that can be operated or controlled by means of an electric current or voltage, a pressurized fluid (for example, a gas or a liquid).

In various embodiments, the actuator **102** can be configured to receive power from a power supply (such as, for example, an electric power supply). The actuator **102** can be configured to receive a control signal to control operation of the actuator, including, for example, to drive or move an actuator drive shaft **311** along an input axis **310** (shown in FIG. 4). The actuator **102** can include an actuator position sensor **312** (shown in FIG. 4), such as, for example, a rotary encoder, that can detect, measure and monitor the position of

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the actuator drive shaft **311** and/or the actuator **102** in real-time, as well as the overall welding system **100**.

The cam **104** can be configured to be adjustable. The cam **104** can be configured to adjust a base line or a starting point or an end point of an energy stack in the ram module **116**. In various embodiments, the cam **104** is configured to be automatically adjustable such as by an actuator (for example, in the driver module **30**, shown in FIG. 1) or manually adjustable, such as, for example, via a cam handle **302**, as seen in FIG. 4.

In certain embodiments comprising the cam actuator (in the driver module **30**, shown in FIG. 1), the cam actuator can include, for example, a motor, a linear servo actuator, a servomotor, a rotary servo motor, a stepper motor coupled to a rotary-to-linear converter, a servomechanism, a linear motor, a stepper motor, a minor loop feedback motor, a pneumatic actuator, a hydraulic actuator, or other drive mechanism configured to transform energy to a linear, rotary, or oscillatory motion, including one or more mechanisms that can be operated or controlled by means of an electric current or voltage, a pressurized fluid (for example, a gas or a liquid). The cam actuator can include a position sensor (not shown) configured to detect and measure the real-time position of the cam **104** and generate a cam position signal and send the cam position signal to the controller **50/200** (shown in FIGS. 1A, 1B).

The frame **106** can be configured to support the actuator **102**, ram module **116**, and the toggle mechanism **120** during each operating cycle, including maintaining positioning and operation of each component according to predetermined specification requirements. The frame **106** is configured to support and hold the actuator **102** as it interacts with or moves (directly or through an intervening device such as the actuator drive shaft **311**) a portion of the toggle mechanism **120**. The frame **106** is also configured to support and hold the ram module **116** as the energy stack in the ram module **116** operates.

The ram module **116** includes a ram **112** and the energy stack. The energy stack includes a transducer **108**, an optional booster **110**, and the device **114**. The ram module **116** can be configured to move the device **114** linearly and press it against the workpiece(s) **60** (shown in FIG. 1) to be treated. As noted above, the workpiece(s) **60** can include one or more materials and/or products to be bonded, welded, soldered, fused, or cut.

FIG. 3 shows a view of the welding system **100** with the ram module **116** removed to show one or more backlash compensation springs **202** (for example, a pair of springs **202a** and **202b**) and a position sensor **204** that can be included, for example, in the thruster assembly TA. In various embodiments, the backlash compensation spring **202** can include any suitable energy transfer mechanism that converts positional displacement to potential energy. The backlash compensation spring **202** (**202a**, **202b**) can be configured to bias any backlash force in a direction of the working load, in parallel with the working load along an output axis **314** (shown in FIG. 4). In some applications, the output axis **314** can be parallel with gravity-specifically, the gravity vector defined by the direction of gravity. The position sensor **204** can be configured to detect, measure, and monitor the position of the ram module **116** in real-time and send a real-time stack position signal to, for example, the controller **50/200** (shown in FIGS. 1A, 1B). The position sensor **204** can include a linear position sensor that detects, measures, and monitors the real-time position of the ultrasonic stack, including the transducer **108**, optional booster **110**, and device **114**.

Linear Encoder

FIG. 4 shows a view of the welding system 100 with the frame 106 removed to facilitate a better understanding of the toggle mechanism 120, according to the principles of the disclosure. The toggle mechanism 120, which comprises (or is linked to) the cam 104 (shown in FIG. 2), includes one or more links that facilitate multi-axial movement of the toggle mechanism 120, such as, for example, in an x-y plane formed by an x-axis and a y-axis. In at least one embodiment, the x-axis includes the input axis 310 and the y-axis includes the output axis 314. The one or more links can include a cam link (or first link) 304, an actuator link (or second link) 307, and a ram link (or third link) 308. The toggle mechanism 120 can be connected to, or incorporate, the force sensor 306, which is configured to detect, measure, and monitor a force applied to any one or more of the links in the toggle mechanism 120, including, for example, the cam link 304, the actuator link 307, and/or the ram link 308, and generate a toggle force signal indicative of the force applied to the particular link(s). The force sensor 306 can be configured to send the toggle force signal via a communication link to, for example, the controller 50/200 (shown in FIGS. 1A, 1B).

As seen in FIG. 4, the toggle mechanism 120 can be attached at one end to the adjustable cam 104 (shown in FIG. 2), which in turn has a portion connected fixedly to the frame 106. The cam 104 is adjustable (automatically or manually) to move the connected end of the toggle mechanism 120 with respect to the frame 106 and, thereby, the welding system 100. The cam 104 can be configured to operate as a fixed pivot of the toggle mechanism 120.

In various embodiments, the toggle mechanism 120 can be connected in the welding system 110 at one or more connection points, including to the cam 104 via the cam link 304. In the embodiment depicted in FIG. 4, the toggle mechanism 120 is connected at three points, including: coupled to the cam 104 via the cam link 304; coupled to the drive shaft 311 and actuator 102 via the actuator link 307; and coupled to a ram drive member 309 via the ram link 308. Specifically, the cam link 304, which includes the force sensor 306 as shown in the illustration on the right side of FIG. 4, is connected at one end to the cam 104 and at another end to the actuator link 307 and ram link 308; the actuator link 307 is connected at a first end to the cam link 306 and ram link 308 and at another end to the actuator drive shaft 311; and, the ram link 308 is connected at one end to the cam link 304 and actuator link 307 and at another end to the ram drive member 309. The ram drive member 309 can be coupled to the ram module 116 along a first side and coupled to the frame 106 along a second side and configured to be parallel to the output axis 314.

In various embodiments, the ram driver member 309 can be configured to travel with respect to the frame 106 or the ram module 116 along the output axis 314.

In certain embodiments, the ram drive member 309 can be connected directly, or through an intervening device (not shown), to the ram module 116 and/or the frame 106. The ram drive member 309 can be configured to move with the ram module 116, or to guide the ram module 116 as it moves, along the output axis 314.

The actuator link 307, which is pivotally coupled to one end of the link 308, can be configured to move along the input axis 310 and thereby force the link 308 to move along the input axis 310 and the output axis 314 and simultaneously pivot about its pivot axis PA since the other end of the link 308 is pivotally connected to the ram drive member 309.

As seen in the embodiment depicted in FIGS. 2-6, the welding system 100 can include a linear position sensor 204 for single (or multi) axis control of an adjustable geometry mechanism where variability in the output axis 314 position is important to setup or operation. During operation, the toggle mechanism 120 can be driven by the actuator 102 along the input axis 310, and the device 114 can move in a linear fashion along the output axis 314. The movement of the device 114 can be detected, measured, and monitored by the position sensor 204 (shown in FIG. 3) in real-time. Simultaneously (or at a different time), the position of the toggle mechanism 120 can be detected, measured, and monitored in real-time by the actuator position sensor 312, which can be integrated with the actuator 102.

In various embodiments, the actuator position sensor 312 can include, for example, a linear position sensor, a rotary encoder, a linear encoder, an optical encoder, or other sensor device capable of detecting and measuring displacement or movement of the drive shaft 311 and/or the actuator 102. The position of the device 114 along the output axis 314 can be determined by the position sensor 204, which in at least one embodiment includes a linear encoder attached to the ram 112 (shown in FIG. 2).

In various embodiments, an optimum operating position of the toggle mechanism 120 for treatment (for example, bonding, welding, soldering, fusing, or cutting) can be such that the cam link 304 and the ram link 308 are in-line, or within manufacturing tolerances to being in-line, with each other with respect to the output axis 314. The cam 104 can be configured and operated to adjust the geometry of the toggle mechanism 120 to accommodate variations in dimensions of the device 114 while maintaining alignment of the cam link 304 and the ram link 308 with respect to the output axis 314. For instance, the toggle mechanism 120 can be used with different types or sizes of horns 114, including horns 114 having different geometries, shapes, or dimensions.

In various embodiments, the controller 50/200 can be configured to monitor and detect unwanted cam movement. In certain embodiments, the controller 50/200 can monitor position signals from both position sensors 204 (shown in FIG. 3) and 312 (shown in FIG. 4) and detect if the cam moved from unwanted instantaneous high force or operator adjustment. If unwanted movement is detected, the controller 50/200 can generate a fault signal and stop machine operation.

Detecting and monitoring the real-time position of the toggle mechanism 120 along the mechanically variable output axis 314 allows for detection and verification of any adjusted geometry of the toggle mechanism 120, which can be outside of the control or direct observation of the welding system 100 or its operator. In various embodiments, the controller 50/200 (shown in FIGS. 1A, 1B) can be configured to receive an input axis position signal from the actuator position sensor 312 and an output axis position signal from the position sensor 204 (shown in FIG. 3). The controller 50/200 can be configured to generate a notification signal and/or to automatically adjust the geometry of the toggle mechanism 120 into alignment with the output axis 314 (shown in FIG. 4). The notification signal can include an alert to an operator indicating, for example, that a part of the machine is out of alignment and/or instructing the operator to adjust the toggle mechanism 120 until it is aligned with the output axis 314.

Before operation of the welding system 100 begins, the real-time position of the toggle mechanism 120 can be determined (for example, by the controller 50/200) to ensure

repeatable production and reduce any line restarts. The toggle mechanism 120 can go out of alignment or otherwise need adjustment, such as, for example, where one or more parts or components in the welding system are accidentally or intentionally adjusted or altered by a user, improperly mounted or installed, become loose or the mounting is or becomes too flexible/compliant, are defective, or become worn, fail, or otherwise cease operating as intended.

By detecting, measuring, or monitoring any change in geometry of the toggle mechanism 120, the welding system 100 can adjust the geometry to ensure proper compensation for external forces or loads. For instance, changes in geometry of the toggle mechanism 120 can be detected and measured by the sensor module 40 (shown in FIG. 1), and the measurements used to adjust the geometry of the toggle mechanism 120, such as, for example, by operating the actuator 102 until the operating axis of the toggle mechanism 120 is parallel to the output axis 314. The operating axis of the toggle mechanism 120 can be parallel to the travel path of the device 114.

For example, referring to FIG. 4, the geometry of the toggle mechanism 120 can be adjusted by the cam 104, such as, for example, by an operator manually moving the cam handle 302 to cause the cam 104 to move one end of the toggle mechanism 120 along the input axis 310 until the operating axis of the toggle mechanism 120 is parallel with the output axis 314.

Referring to FIG. 3, each backlash compensation spring 202 (202a, 202b) can be configured to bias backlash in the direction of the working load, in parallel with the working load component along the output axis 314. The force exerted (F Spring) by each backlash compensation spring 202 can be dependent on the linear position of one (or both) ends of the spring, such as, for example, with respect to the frame 106 or the ram 112. Accordingly, when measuring a load (for example, by the force sensor 306) at the cam link 304 and the ram link 308, the measured load is subject to the force exerted by the spring 202, which can be subtracted from the measured load, for example, by the controller 50/200 (shown in FIGS. 1A, 1B) to calculate the actual external load.

The controller 50/200 can be configured to calculate the actual external load by, for example, calculating the sum of all x-vector forces and all y-vector forces detected and measured by the force sensor 306 and subtracting the sum of y-vector backlash spring forces, which in the embodiment depicted in FIG. 3 is $2 \cdot F_{spring}$ since the backlash springs 202a, 202b are substantially the same and oriented so as to apply the backlash force along the output (or y-) axis 314.

Referring to FIG. 4, the force sensor 306 can be configured to detect and measure a force (F) comprising an x-vector force component (F_x) and a y-vector force component (F_y) that are transferred to the force sensor 306 via the actuator link 307 and/or the ram link 308. In an embodiment, the controller 50/200 (shown in FIGS. 1A, 1B) can calculate an external load (L_{ext}) by the following equation (1):

$$L_{ext} = F - 2 \cdot F_{spring} \quad (1)$$

In many applications, the welding system 100 will be configured such that any external load (L_{ext}) will have substantially only a y-vector force component (L_y)—that is, the load will be substantially entirely along the output axis 314 (shown in FIG. 4). Since the force exerted by the two backlash springs mostly has only a y-axis component, the external load L_{ext} can be calculated by the following equation (2):

$$L_{ext} = L_y - 2 \cdot F_{spring} \quad (2)$$

Since the force F and load L are transferred to the actuator link 307 and/or the ram link 308, the force sensor 306 can be positioned near the actuator link 307 and/or ram link 308, such as, for example, within the cam link 304 (as seen in FIG. 4), where the aggregate of all forces and loads exerted on the toggle mechanism 120 can be measured by the force sensor 306. Thus, when measuring (by the force sensor 306) an external load L_{ext} that is applied to, for example, the device 114, forces or loads experienced internally such as, for example the force exerted by each backlash spring 202 must be removed or compensated for to get an accurate measurement.

Referring to FIG. 3, the position sensor 204 can be configured to detect and measure the real-time position of the toggle mechanism 120 and/or energy stack (including the device 114) along the output axis (or y-axis) 314 at any point in time and generate an output axis position signal. The actuator position sensor 312 (shown in FIG. 4) can be configured to detect and measure the real-time position of the toggle mechanism 120 and/or energy stack (including the device 114) along the input axis (or x-axis) 310, at the same or a different time as the position is measured by the position sensor 204. Based on the real-time output axis position signal from the position sensor 204 and the real-time input axis position signal from the actuator position sensor 312, the real-time geometry of the toggle mechanism 120 can be detected and the sensor signals used to determine an adjustment value for the cam 102 to align the toggle mechanism with the output axis 314.

An effective mechanical advantage of the toggle mechanism 120 can be dependent upon a narrow window of operating positions of the toggle mechanism 120 and, thereby, the energy stack, including the device 114. The toggle mechanism 120 (and thereby the energy stack) can be maintained within the narrow window of operating positions by monitoring the output axis position signal and the input axis position signal and adjusting the toggle mechanism 120 when it exceeds either end of the window of operating positions. In certain embodiments, the window of operating positions can be set to, for example, a linear operating range of 600 μm to 1000 μm within the max obtainable linear position to ensure proper operation of the welding system 100. Thus, when the geometry of the toggle mechanism 120 is determined to exceed either end of the window of operating positions, the cam 104 can be operated to adjust the geometry and return the toggle mechanism 120 to a default position, in which the operating axis of the toggle mechanism 120 is substantially parallel to the output axis 314.

The window of operating positions can be exceeded at either of its ends in real world applications of the welding system 100. For example, a violation of the minimum threshold can result in an unnecessary force being transmitted from the output axis 314 to the input axis 310 and the force being poorly translated into the link(s) that contain the force sensor 306 (shown in FIG. 4); and a violation of the maximum threshold can occur such as in “over center” conditions where increased positions along the output axis 314 positions (and therefore working force) cannot be obtained or controlled by the welding system 100 due to the inversion of the toggle mechanism 120, where the direction of motion of the output axis becomes opposite of the direction of the motion of the input axis. By monitoring the first and second position sensors, the inversion point can be identified when positive motion (advancement) on the input axis results in negative motion (retraction) on the output axis. Both violations indicate that the ram 112 is out of alignment and that mechanical adjustment of the toggle

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mechanism 120 is necessary, with the violation of the maximum threshold possibly indicating that mechanical adjustment of the toggle mechanism 120 has become necessary, for example, due to component wear. In various embodiments the controller 50/200 is configured to generate an alarm when the travel path exceeds a predetermined threshold, such as, for example, when the toggle mechanism is beyond its inversion point or the travel path is 600 μm , 1000 μm , or greater, past full stroke of the welding device with respect to the input axis.

Real-time sensor data representing the linear position or displacement of the ram 112 during each treatment (for example, a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation) can identify out-of-specification operation not observable from the force sensor signal alone, such as, for example, the real-time sensor data signal from the force sensor 306 (shown in FIG. 3). This can happen, for example, due to component wear or replacement, a change in processed or raw material, mechanical adjustment, device settings, or any combination thereof. As a result, detection and measurement of the position and movement of the ram 112 by the position sensor 204 can be used to ensure consistency.

Additionally, where the welding system 100 is included in two or more machines, the combination of the force signal from the force sensor 306 and the position signal from the position sensor 204 can be used to maintain consistency in operation from machine to machine by maintaining the toggle mechanism 120 and, thereby, the ram 112 in proper alignment, even though variations exist between the machines, such as, for example, due to manufacturing tolerances or misalignment resulting from improper shipping, assembly or installation. The welding system 100 is configured to provide optimal performance, including under varying operating conditions or when switching products (for example, to be bonded, welded, soldered, fused, or cut), by ensuring alignment of the ram 112 and device 114 with the output axis.

The present disclosure also contemplates precision positioning during setup using user definable linear offsets. The "Goto Stack Alignment Position" feature will extend the ram to a user fixed linear offset, which is just short of (for example, by 100 μm) the known maximum displacement. This represents an optimized operating position, and once the ram is in this position mechanical alignment can be done with repeatable results. The "Find Dynamic Ready Position" feature will extend the ram until a force signal is encountered representing the displacement needed to reach the operating position given the current mechanical alignment. The ram will then retract a user definable linear offset from that position, providing consistent timing at weld cycle start up.

In various embodiments, the actuator 102 (including sensor 312) can include a combination of a servomotor with encoder feedback and the position sensor 204 can include a linear encoder. In those embodiments, the welding system 100 can monitor the relationship of the actuator position sensor 312 to the linear encoder 204 to detect and measure alignment of the ram 112. If there are any changes outside of a predetermined band, such as, for example, the toggle mechanism 120 (or ram 112) being out of alignment enough to cause a difference in the expected linear position with respect to the output axis 314 (shown in FIG. 4), then a fault can be detected or determined and an alert signal generated and an alert issued to the operator to indicate that the actuator 102 is out of adjustment and needs to be adjusted. This can happen, for example, when rogue material that's

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too thick is presented between the device 114 and an anvil 115 (shown in FIGS. 13A, 13B).

The welding system 100 can also detect overtravel of the actuator 102 and/or the actuator drive shaft 311. For example, referring to FIG. 4, if the actuator link 307 is extended too far, the linkage will invert, which may keep the drive shaft 311 moving but not move the ram 112 any further. State-of-the-art systems can run past the linkage inversion point and become damaged, whereas the weld system 100 has a safety function with overtravel detection via the actuator position sensor 312 and/or the position sensor 204 and a process for overtravel detection and alert signal generation, issuing an alert of a fault condition such as, for example, the actuator 102 or cam 104 being out of adjustment. The welding system 100 is configured to provide detection of actual linear displacement of the device 114 and generation of an alert signal thereof, which allows the welding system 100 to inform the operator when a component (for example, a linkage) in the toggle mechanism 120 needs to be repaired or replaced, or when overtravel is detected.

In various embodiments, the alignment adjustment and correction of the toggle mechanism 120 can be automated, in which case the controller 50/200 can be configured to receive the various sensor signals, including the input axis (or x-axis) position signal from the actuator position sensor 312 and/or the output axis (or y-axis) position signal from the position sensor 204, and generate a cam adjustment drive signal to drive the adjustable cam 104 to a point at which the toggle mechanism 120 (or ram 112) is substantially parallel with the output axis 314.

The system 100 is configured to monitor operation of its various components, including the actuator 102 and the toggle mechanism 120, and when a predetermined condition occurs, such as, for example, a predetermined number of mils/micrometers of mechanical play is noticed in the linkages (for example, cam link 304, actuator link 307, ram link 308, or a combination thereof), the system 100 can generate and provide a shutdown warning or alarm that the machine needs to be serviced. The operator can be notified that maintenance is required or advised to shut down the machine.

In various embodiments in which the geometry adjustment is automated, the controller 50/200 can be configured to monitor operation of the various components in the welding system 100, including the actuator 102 and the toggle mechanism 120, and when a predetermined condition occurs, such as, for example, the predetermined number of mils/micrometers of play is noticed in the linkages, the controller 50/200 can stop operation, generate and provide a shutdown warning or alarm that the machine needs to be serviced, and automatically adjust the actuator 102 and/or the toggle mechanism 120 to adjust the geometry into alignment, for example, by aligning the toggle mechanism 120 with respect to the output axis 314. The operator can be notified that maintenance is required or advised regarding the status of the maintenance as it is performed automatically. In certain embodiments the controller 50/200 can be configured to hold the actuator 102 at a linear position that is near metal-to-metal contact, such as, for example, between 10 μm and 1000 μm of having metal-to-metal contact, or 100 μm or less in at least one embodiment.

In various embodiments, the force sensor 306 (shown in FIG. 4) might require periodic re-calibration. The welding system 100 can be configured to perform force calibration for the full force range of the sensor without requiring the removal of the thruster assembly TA from the machine, or

the removal of the device **114** from the ram **112**. Without this configuration of the system **100**, it would be necessary to remove the thruster assembly TA from the machine to calibrate the force sensor **306**, which is a time-consuming and labor-intensive task. Compared to the state-of-the-art ultrasonic welding machines, the welding system **100** provides a significant reduction in the time to perform calibration of a force sensor, as well as elimination of any need to re-level the device **114** once calibration is complete.

FIGS. **5** and **6** show different views of a calibration system that can be included in the welding system **100**. FIG. **5** shows a view of the welding system **100** including a pair of calibration protrusions **402** (**402a**, **402b**) attached to or formed as part of the frame **106**. The protrusions **402a**, **402b** can include, for example, shafts, half shafts, pins, rods, clips, screws, bolts, nuts, or the like. In the embodiment depicted in FIG. **5**, the protrusions **402a**, **402b** each include a half shaft. The transducer **108** is removed/hidden in FIG. **5** for ease of illustration.

FIG. **6** shows a tension block **502** and a calibration load cell **504** attached between the frame **106** and the booster **110**. The tension block **502** can have any shape that facilitates easy and secure installation between the frame **106** and the booster **110** (or device **114**), with the calibration load cell **504** installed between the tension block **502** and the booster **110** (or device **114**).

Referring to FIGS. **5** and **6**, a force calibration process can be performed by first removing the transducer **108** from the thruster assembly TA. The tension block **502** can then be placed over the half shafts **402a**, **402b**, and the calibration load cell **504** can be attached between the tension block **502** and the booster **110** (or the device **114**) by attaching an end of the calibration load cell **504** to a portion of the tension block **502** and an opposite end of the calibration load cell **504** to a portion of the booster **110** (or the device **114**). The toggle mechanism **120** can then be extended to a predetermined position, such as, for example, the same position of the toggle mechanism **120** during operation. As a force **F** is applied along the input axis **310** (shown in FIG. **4**), a compressive load is produced in the force sensor **306** (shown in FIG. **4**) and a tension load is produced on the calibration load cell **504**. The force signals from the force sensor **306** and the calibration load cell **504** can be sampled for different input axis forces and a table populated in the controller **50/200**, which then correlates the calibration load cell force with the sensor force. During operation, the force applied by the device **114** is determined by referencing the sensor force readings against this calibration table stored in the controller **50/200**, including interpolation between table points.

In at least one embodiment, the controller **50/200** (shown in FIGS. **1A**, **1B**) is configured to execute instructions to provide for metal contact detection and alert signal generation, including generating a warning signal when there is metal contact and sending (for example, via the IO interface **230** or communication unit **240**) the warning signal to an output device (not shown) such as a human-machine interface (HMI) (for example, a touchscreen display device) to display a warning to an operator. If the warning or a set of warnings is ignored or not resolved and the metal contact worsens, then a fault can be determined by the controller **50/200** and a fault signal can be generated and sent to the output device (not shown). Using the HMI, an operator can interact with the controller **200** to set limits on when a warning condition or a fault condition is determined, including to set error limits. The controller **50/200** can be configured to receive commands, via the HMI, including to set a warning setpoint such as, for example, percentage of set-

point deviation for ultrasonic amplitude, ultrasonic frequency, ultrasonic lock quality, or mechanical position. By monitoring ultrasonic frequency output of the ultrasonic generator (not shown), its Standard Deviation (StD) could be calculated and monitored. Once a metal-to-metal contact occurs, StD value will increase abruptly. Specific value of the frequency StD is applications dependent-horn size, speed, anvil details, non-woven materials properties and force all affect such values. For most applications the increase in StD is expected to be in 10%-200% range, or greater.

Frequency Lock Quality is a measure of the volatility of the frequency regulation function of the welding system **100**. It is an error signal representing how accurately the synthesized output frequency matches the ultrasonic feedback. The frequency regulation works to minimize this value. Elevated levels of error, either temporary or sustained, indicate a mismatch between the control response and environment. For a given frequency regulation tuning and operating environment, Lock Quality will be largely consistent and predictable. The differences in these Lock Quality values for known operating environments, for example, with or without the presence of horn-to-anvil contact, can be compared to determine which operating environment is currently present. For Frequency Lock Quality its Standard Deviation value increase could also be used to detect metal-to-metal contact. For most applications the increase in Frequency Lock Quality StD is expected to be in 10%-200% range, or greater.

In certain embodiments, the weld system **100** can include a controller **200** configured for a sub state of a weld operation that provides for rapid position change to create space for a splice in material. To perform such a quick move, the controller **200** can be configured with a separate set of tuning parameters, and upon completion of the movement immediately re-enter the weld state using state information from before the splice move started, including BBT and ACAR adaptations. This minimizes the wasted parts associated with a material splice. The controller **200** equipped with BBT is aware of the splice move state and pauses the bond detection operation (such as, for example, the process **900**, shown in FIG. **10**) while in a known retracted unloaded position. The splice move can be synchronized to the inter-bond idle period. The controller **200** can be configured to receive user programmable splice distance inputs, which can be specified in linear encoder units, and provide predictable/measurable minimized nip gaps for the thicker splice material to pass.

Bond Balance Technology (BBT)

The welding system **100** (or workstation **1**, shown in FIG. **1A**) can be configured for characterization of individual bonds of a dual bond anvil pattern in an intermittent bonding application using no external anvil position/angle feedback sensors. As discussed above with reference to FIG. **1A**, the welding system **100** can include first and second treatment modules **10**, **20**, either of which can include a welding device such as the device **114** or an anvil **115** (for example, shown in FIG. **13A**, or **13B**). In various embodiments, the welding system **100** can include the Bond Balance Technology (BBT) that can be particularly useful for short-time intermittent bonding applications, as well as for continuous bonding applications.

In the welding system **100**, the linear operating position of the ram **112** can be used to adjust an applied force of the device **114**. For example, to adjust a linear operating position, the device **114** maybe backed away from the anvil **115**

slightly to reduce a noise signal. The device **114** can be backed off until metal (horn-anvil) contact goes away.

FIGS. **13A** and **13B** show two nonlimiting embodiments of a horn-anvil combination that can be included in the welding system **100** (or workstation 1, shown in FIG. **1A**) for treating a workpiece **60**. As seen, the anvil can include a rotary anvil **115** having one or more welding features **115-1**, **115-2**, . . . , or **115-n**, where n is a positive integer greater than 2. Each welding feature **115-n** can include a contact portion, a row, a column, or a ridge having notches and lands or protrusions, as understood by those skilled in the art. Each welding feature **115-n** can be configured to contact with the device **114** to form a bond.

The welding system **100** can be configured to automatically back off a position by increasing the distance of the anvil **115** from the device **114** based on a signal indicating metal contact. In various embodiments, metal contact between the device **114** and the anvil **115** can be detected based on, for example, the force signals received from the force sensor **306** (shown in FIG. **4**) and/or the linear position sensor **204** (shown in FIG. **3**) without using any external metal conductivity sensors at the device **114** or the anvil **115**.

In various embodiments, the welding system **100** (or workstation 1) is configured to process operational data, including peak force, average force, bond period and duration, and ultrasonic power/energy. The welding system **100** can control bond force toward a selectable reference/feedback force of overall bond peak force or the average of the peak forces of the individual bonds for intermittent applications, and variable force for continuous applications. The welding system **100** can provide transparent operation between anvil designs with any number of identical bonds per rotation without adjustment of any manual settings, supporting any bond duty cycle. The welding system **100** features automatic built-in cycle rate detection without requiring any external sensors.

FIGS. **7A** and **7B** show two different tensile strength plots from trailing to leading edges without BBT operations (FIG. **7A**) and with BBT operations (FIG. **7B**) performed by the welding system **100**, for example, equipped with the anvil **115** with two welding features **115-1**, **115-2** (shown in FIG.

13A) for illustration purposes. In various embodiments the trailing edge **602/612** corresponds to a trailing edge portion of a workpiece **60** (or unit of a workpiece) and the leading edge **604/614** corresponds to a leading edge of adjoining workpiece **60** (or unit of the workpiece), with a process cycle rate (or production speed) of, for example, 150, 175, 200, or more workpieces (or units of workpieces) per minute, including welding, bonding, fusing, and/or cutting. As can be seen in FIG. **7A**, the bond strength variability between trailing-leading portions of workpieces **60** can be high, wherein the tensile strength of a trailing edge **602** of a bond is significantly higher than a corresponding tensile strength of a leading edge **604**—for example, a pull strength difference of about 3 Newtons/inch (N/in.). However, with the BBT operations performed, as seen in FIG. **7B**, the bond strength variability is significantly reduced, wherein the tensile strengths of a trailing bond **612** versus a leading bond **614** are nearly the same. As seen in FIGS. **7A** and **7B**, by performing BBT operations (for example, via the controller **50**) bond strength variability can be reduced significantly. For instance, before a BBT operation (FIG. **7A**), the bond strength (for example, as measured by pull strength) of the trailing edge **602** of a workpiece **60** can be, for example, between 11.0 and 16.4 Newtons-per-inch (N/in.) and the leading edge **604** of the adjoining workpiece **60** can be, for example, between 7.2 and 10.8 N/in.; and with the BBT operation ON (FIG. **7B**) the bond strength variability is reduced such that the bond strengths of the trailing edge **612** and leading edge **614** are similar, for example, the bond strength (for example, as measured by pull strength) of the trailing edge **612** of the workpiece **60** can be, for example, between 13.0 and 17.2 N/in. and that of the leading edge **614** of the adjoining workpiece **60** can be, for example, between 9.8 and 17.6 N/in. It is noted that an “edge” also means a bond and the terms are used interchangeably.

Table 1 below illustrates some example outcomes for BBT operations OFF (or deactivated) versus BBT operations ON (or activated). An 86% improvement can be seen in bond consistency when BBT operations are turned on compared to a state-of-the-art process.

TABLE 1

Settings Used	Amplitude	Speed (ppm)	Speed (mpm)	Force (N)	BBT
Code 1	100%	250	150	3000	OFF
Code 2	100%	250	150	3000	ON
Code 3	100%	250	150	3000	OFF
Code 4	100%	250	150	3000	ON

Code 1			Code 3		
	Trail (N/in.)	Lead (N/in.)		Trail (N/in.)	Lead (N/in.)
REP 1	13.6	9	REP 1	14.4	9.8
REP 2	13	9.2	REP 2	13.4	9
REP 3	12	9.4	REP 3	12.8	7.2
REP 4	13.6	9.4	REP 4	12.4	8
REP 5	12.4	9.6	REP 5	17.2	8.6
REP 6	11	8.4	REP 6	13	7.8
REP 7	11.4	8.8	REP 7	16.4	8
REP 8	14.8	10.8	REP 8	14	8.2
REP 9	9.4	7.4	REP 9	14.2	10
REP 10	13	10.8	REP 10	15.4	8.8
AVG	12.42	9.28	AVG	14.32	8.54
DIFF.		25.28%			40.36%

CODE 2			CODE 4		
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TABLE 1-continued

	Trail (N/in.)	Lead (N/in.)		Trail (N/in.)	Lead (N/in.)
REP 1	16.6	15.2	REP 1	15.2	12
REP 2	16.4	17.6	REP 2	16.6	14.8
REP 3	15.6	15	REP 3	14.6	14.4
REP 4	14.2	14.6	REP 4	14.2	14.2
REP 5	17.2	17.2	REP 5	16.8	9.8
REP 6	16.8	17	REP 6	16.2	16.6
REP 7	16	15.8	REP 7	13.8	12.2
REP 8	14.2	16.4	REP 8	13	12.6
REP 9	13	15.6	REP 9	17.2	13.2
REP 10	13.6	14.8	REP 10	14.8	13.2
AVG	15.36	15.92	AVG	15.24	13.30
DIFF.	-3.65%				12.73%
	IMPROVEMENT	86.16%			

In various embodiments, the BBT operation (for example, performed by the controller **50/200**) includes: receiving a force signal at each of the trailing edge **602/612** and the leading edge **604/612** of the workpiece(s) **60**; determining (for example, by the controller **50/200**) the force on the treatment module **10** (or **20**, shown in FIG. 1), including the peak force value for the trailing edge **602/612** and the peak force value for the leading edge **604/614**; determining an average of the two peak force values; determining an operation adjustment value for the treatment module **10** (or **20**); and automatically applying (for example, by the controller **50/200**) the operation adjustment value to increase, decrease or maintain the force applied to the workpiece(s) **60**. The operation adjustment value can include, for example, an amount of force to be applied to the workpiece(s) **60**.

FIG. 10 shows an embodiment of a process **900** for detecting bond properties during an ultrasonic welding operation. The process **900** can be performed by, for example, the welding system **100** equipped with the controller **50/200** (shown in FIGS. 1A, 1B).

Initially, the process **900** starts in an idle state (Step **902**) and runs short and long averaging of the force signal in the idle state (Step **904**). Short averaging periods are on the order of single digit milliseconds, and long averaging can range from 100's of milliseconds to multiple seconds. Averaging time can be chosen empirically based on specific application needs (speed, force, material properties) to optimize the weld quality. When a bond cycle starts, a force signal is tested for the bond start using a user-defined threshold (Step **906**). If a bond is detected, the bond start time is logged and recorded for bond period calculation, and a bond status bit is set (Step **910**), otherwise the process continues to run short and long force averaging (Step **904**).

During the bond duration, peak forces can be detected and recorded for bond balance (Step **912**). When the bond has been completed, the state moves to a process bond state in which the bond status bit is cleared, the bond stop time is logged and recorded for bond duration calculation, bond statistic running averages are updated, and bond treatment period and duration are updated for the detection of the treatment position in time (Step **914**). The process returns to a bond detect state to test the force signal for the bond start using bond balance data threshold (for example, instead of a user-defined threshold) (Step **908**).

FIG. 9 shows the force signal **804** of a typical anvil pattern preforming two bonds per cycle. One can see an imbalance in the peak force values of the two anvil contact points **806** and **809**. Existing embodiments only sample the overall peak and regulate to that maximum, in this case **809**. Bond

Balance Technology (or BBT) recognizes the individual peaks, records their values, and averages one or more of these values to determine the force regulation reference value.

In the embodiments of the welding system **100** (or workstation 1) equipped with BBT, the system **100** (or workstation 1) can be configured to perform the process **1100** seen in FIG. 12. During process **1100** can be performed, for example, in an intermittent bonding application or a continuous bonding application. Referring to FIGS. 12 and 13 together, the controller **200** (or **50**) is configured to receive a force signal and detect (for example, by the processor **210**) for a start of a bond based on the force signal (Step **1105**). If a bond start is detected (YES at Step **1105**), then the force signal is analyzed (for example, by the processor **210**) for local maximums, including each bond force peak in an average of bond force peaks (Step **1110**); otherwise (NO at Step **1105**) the process continues to monitor and detect for bond starts (**1105**). The results of the force signal analysis (Step **1110**) are monitored (for example, by the processor **210**) to detect the end of the bond (Step **1115**).

If bond end is detected (YES at Step **1115**), then the average of the bond force peaks encountered during the bond can be calculated, for example, by the processor **210** (Step **1120**); otherwise (NO at Step **1115**) the process continues to monitor the analyzed force signal (Step **1110**) to detect a bond end. The average of the bond force peaks can be included in a running average of previous results to form a BB reference force (Step **1120**). In various embodiments the controller **200** (or **50**) is configured to detect and recognize each individual bond force peak (for example, by the processor **210**), record a value for each bond force peak, and average two or more of the values to determine a force regulation, or position regulation reference value.

A force regulation error can be calculated (for example, by the processor **210**) based on the BBT reference force (Step **1125**) by comparing a predetermined target force (for example, preset by a user) and the BBT reference force value. Based on the calculated force regulation error, an operating condition can be determined (for example, by the processor **210**), such as whether the force regulation error is greater than a predetermined threshold value, for example, preset by the user (Step **1130**). If an operating condition is determined in which the force regulation error is greater than the threshold value (YES at Step **1130**), then the servo position of the treatment module **10** (or **20**, shown in FIG. 1) can be adjusted to minimize the force regulation error (Step **1135**); otherwise (NO at Step **1130**) the servo position can be maintained (Step **1140**). The process **1100** can continue and repeat Steps **1105** to **1140** for each bond.

In the course of characterizing and collecting data for the bond force peak average calculation, a bond force span value is also calculated as the difference in the minimum and maximum peak values encountered during the treatment. This signal serves as a bond balance error signal to be monitored for process consistency.

As evident from the above descriptions, advantages of the welding system and processes according to the instant disclosure include, but are not limited to: bond strength variability less than 10%; both sides (leading and trailing) of a bond are near equal, no more weak or uneven bonds; faster sampling rates, taking an average of peaks instead of sampling to peak force; automatic cycle rate detection doesn't require external cycle rate feedback; actuator out of adjustment detection and correction using dual encoder feedback; actuator overtravel detection; automatic maintenance required warning for mechanical actuation linkage; actual linear displacement of the horn; and specific embodiments configured for intermittent bonding applications; metal contact detection without any need for external sensors.

Adaptive Cyclic Amplitude Regulation (ACAR)

In various embodiments, the welding system **100** can include Adaptive Cyclic Amplitude Regulation (ACAR). In at least one embodiment, the controller **200** (shown in FIG. 1B) is configured to execute instructions to perform ACAR operations as described herein.

ACAR provides improved ultrasonic amplitude regulation, and power delivery, to the workpiece **60** (shown in FIG. 1A, 13A, or 13B), during the abrupt introduction, and throughout the duration, of heavy ultrasonic loads encountered during repetitive intermittent bonding applications with a physically fixed bond duty cycle determined by anvil geometry. This works toward reducing ultrasonic amplitude sag, such as, for example, might be associated with near instantaneous changes in load, by using the regulation output of previous bonds to predictively anticipate the current bond. State-of-the-art control techniques such as PID (proportional-integral-derivative) only retroactively respond to the feedback signal. Such state-of-the-art approaches, however, are not suitable for high bond rate or high load applications with short bond durations. In various embodiments, the welding system **100** uses distinctly different control techniques during the unloaded or idle sonics period between bonds and the bonds themselves.

In the embodiments of the welding system **100** (or workstation 1, shown in FIG. 1A) equipped with ACAR, the system can be configured to adapt to the ultrasonic load encountered during individual bonds by, for example, the processor **210** (shown in FIG. 1B) analyzing the system's performance from previous bonds. The controller **50/200** can continuously adapt to the shape of the load (for example, workpiece **60**, shown in FIG. 1A, 13A, or 13B), applying more or less excitation energy throughout the following or subsequent welds. The controller **50/200** can be configured to accommodate bond patterns of any shape or number of anvil contact points per bond, such as, for example, the welding features **115-1** to **115-n**, shown in FIGS. 13A and 13B.

The welding system **100** can include one or more "non-sonic" device sensors (not shown), the outputs of which can be used by the controller **200** to determine the position, period, and duration of the bond, or closely grouped set of bonds. In the embodiment depicted in FIGS. 2-6, the controller **200** can use the force signal received from the force sensor **306** (shown in FIG. 4). In various embodiments, the welding system **100** can include and use linear position and/or ultrasonic power. The welding system **100** can be

configured to data acquisition rates that far exceed update rates of state-of-the-art networked machine controllers (motion control protocols/controllers excluded). This aids in identifying machine variability not visible to existing systems. This variability can be present in advanced machine designs using festooning techniques to protract the bond duration, or "multi-up" anvil designs with multiple bond landing locations.

In various embodiments, immediately before a bond is expected by the controller **200**, there is a short period of "pre-boost" that ramps up ultrasonic amplitude in anticipation of the ultrasonic load step function. This counteracts the loss of inertia/momentum in the energy stack comprising the device **114** that otherwise cannot be accounted for due to physical system frequency response to changes in excitation energy. Ultrasonic amplitude can be measured and monitored by the controller **200** to ensure safe operating conditions for the stack are maintained during this short undamped period of time. The controller **50/200** can anticipate, for this phase of ACAR control, small changes/jitter in the timing of the start of the bond, wherein ACAR control can enter the bond phase early if it is detected earlier than expected. ACAR control can have a maximum "boost" level that it will maintain for a short period if the bond/load arrives later than expected. ACAR implementation eliminates nuisance overloads by reducing amplitude peaks (voltage overloads), which increases equipment reliability and increases transducers lifetime. It also increases the practical range of bonding forces. ACAR implementation also improves bonding of thin non-woven materials that are more sensitive to over-welding due to unwanted amplitude spikes.

During the bond, many/multiple areas of amplitude regulation can be established and distinctly controlled from each other by the controller **50/200**. These areas can be automatically reduced or expanded by the controller **50/200** to match the bond duration as line rates change.

Similar to pre-boost, the ACAR control can be configured for a "post boost" phase that reduces amplitude overshoot as the bond finishes and the large ultrasonic load/damping suddenly disappears. Excitation energy is removed proactively allowing resonant mechanical inertia to dissipate into the workpiece **60** (shown in FIG. 1A, 13A, or 13B) and complete the bond.

The controller **50/200** configured with ACAR automatically adapts to ultrasonic load changes due to line rate changes, target weld force changes, and/or material changes without user interaction, using no external sensors or machine state information. The controller **50/200** can optionally provide bond rate information to the welding system **100**. This improves line rate change response during high levels of acceleration and/or deceleration, provided the controller **50/200** information update rate is acceptable and accurate. The controller **50/200** can be configured to compare/cross check the information updates to its own analysis of the machine state. Alternatively, the controller **50/200** can be configured to operate stand alone, without information updates if unavailable due to legacy machine constraints or other conditions.

In an embodiment, the controller **50/200** can be configured with ACAR intended for continuous pattern rotating anvils, or applications where the material traverses a non-rotating anvil, which do not exhibit large changes in force per treatment. As discussed above, with reference to FIG. 1A, the first treatment module **10** can include a rotating anvil (for example, anvil **115**, shown in FIGS. 13A, 13B) and the second treatment module **20** can include an ultrasonic horn (for example, **114**, shown in FIGS. 13A, 13B). This ACAR

solution could be also applied, for example, to other continuous applications where the ultrasonic stack or material is traversing, or other applications that do not require a rotating anvil. In this embodiment, the controller **50/200** can use changes in ultrasonic amplitude to control the areas of bonded and unbonded materials, such as, for example, the multilayer nonwoven materials. Using ACAR, the controller **200** can provide distinct control tunings for low/idle amplitude unbonded areas, high amplitude bonded areas, and any transitions between them, to ensure rapid convergence on a regulated state in each stage. When line synchronization to the material is not needed, user settings of bond period, duration, unbonded amplitude target, and bonded amplitude target can be used by the controller **200** to allow the system **100** to function with no external inputs. When synchronization is desirable, the system **100**, under control of the controller **200**, will react to an external “bond control” signal received by the controller **200**. The bond control signal can include either a digital input signal or analog signal representing a target amplitude.

In an embodiment, the controller **50/200** can be configured with ACAR to provide varying amplitude targets for each bond of the treatment operation to account for variations encountered in the plurality of bond force characteristics. For example, the amplitude target for bond X can be 80%, and the amplitude target for bond X+1 can be 90%.

While the above embodiments refer to amplitude, the present disclosure contemplates that instead of using amplitude, the power signal can be used.

In FIG. **8A**, the plot diagrams **700** illustrate instantaneous curves of the ultrasound amplitude **702**, bond status **708**, ultrasonic excitation energy **710**, and force **718**, before, during, and after a bond, with a bond duration being, for example, 15 mS, with ACAR disabled. The ultrasonic excitation energy **710** can include a pulse-width, an electronic output effort, or an amplitude regulation control output response. The bond status curve is derived from the force signal **718**. The ultrasonic excitation energy **710** represents the relative output effort of the ultrasonic driver in the treatment module **10** (or **20**, shown in FIG. **1A**), which in the depicted embodiment is a pulse-width. As seen, the amplitude **702** of ultrasonic vibration dips from 100% unloaded (before the bond) to 74.5% during the bond before overshooting to 115%, a 40.6% swing. In various embodiments the bond status **708** includes a square wave pulse during which, for example, the first treatment module **10** and/or the second treatment module **20** (shown in FIG. **1**) contact the workpiece **60** to perform a bonding, welding, fusing, and/or cutting operation.

By comparison, FIG. **8B** illustrates plots **720** in which ACAR has been enabled with dramatic results. Here, the amplitude dips from 100% unloaded to 96.7% with a max of 106%, reducing the range to a 9.3% swing.

With ACAR disabled, the ultrasonic pulse-width is seen to increase, as indicated by **712** in response to the sag in amplitude **704**. The traditional PID control builds output level attempting to reduce the error in the amplitude signal. Once the bond has passed and the load/damping to the ultrasonic device **114** is removed, the ultrasonic excitation energy/output energy is now too large, as indicated by **714**, and the amplitude responds by rising past the intended operating point **706**. With ACAR enabled, the pulse-width is preemptively increased to match the expected load with pre-boost **732** of the pulse-width, adapts throughout the leading and trailing portions of the bond **734**, and switches back to idle/no-load regulation **736** after the bond, minimizing the disturbance to amplitude **722**.

Because the bond occurs within milliseconds, there is a transient response as the bonds traverse the surface of the welding module, such as, for example, an ultrasonic device **114**.

In various embodiments the welding system **100** (or **1** shown in FIG. **1**) includes a welding device **114** for treating one or more workpieces **60** under operation of the controller **50/200**. The controller **50/200** is arranged to perform an ACAR operation, including: receiving an amplitude signal indicative of a real-time amplitude of ultrasonic vibration of the joining device **114** during a treatment cycle; segmenting the amplitude signal into a plurality of amplitude segments for the treatment cycle; monitoring an amplitude value of each of the plurality of amplitude segments during the treatment cycle; employing a plurality of amplitude segments with individual closed-loop control algorithms to calculate a plurality of amplitude adjustment values, each of the plurality of amplitude adjustment values corresponding to a respective one of the plurality of amplitude segments; and applying each amplitude adjustment value to the corresponding respective one of the plurality of amplitude segments during the subsequent treatment cycle in real-time, thereby applying what is detected on a present bond to the next bond, or subsequent (or next) treatment.

In various embodiments each individual closed-loop control algorithm can include a separate PID tuning loop generated for, and corresponding to, one of the plurality of amplitude segments, such that a plurality of distinct PID tuning loops are generated and employed for corresponding amplitude segments during a single pulse **708/728** (shown in FIGS. **8A**, **8B**). The treatment cycle comprises treating a single workpiece **60** in a plurality of workpieces **60**, and the treating comprises a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation. The controller **50/200** can be arranged to generate a proportional-integral-derivative (PID) setting for each of the plurality of amplitude segments and apply each PID setting to the respective one of the plurality of amplitude segments. In various embodiments the controller **50/200** includes a PID control module configured to generate a PID setting (for example, optimal PID gain value) for each amplitude segment to regulate an ultrasonic welding process by adjusting the power applied for the corresponding segment based on real-time measurements of, for example, an amplitude signal, a force signal, a power signal, and/or one or more position signals.

The controller **50/200** can be arranged to receive and analyze a force signal from the force sensor and detect a treating cycle rate based on a timing analysis of the force signal. In certain embodiments the controller **50/200** can be configured to receive and analyze the amplitude signal from amplitude sensor, the power signal from a power sensor and one or more position signals from at least one position sensor, any of which can be included, for example, in the sensor module **40** (shown in FIG. **1A**). The treating cycle rate comprises a rate (in units-per-minute) at which each workpiece **60** in a plurality of workpieces (**60**) is treated, wherein the treatment can include a welding operation, a bonding operation, a cutting operation, or other processing of the workpieces **60**.

FIG. **11A** illustrates an embodiment of a process **800** of the welding device **100** (or workstation **1**) equipped with ACAR, according to the principles of the disclosure. Referring to FIGS. **11A** and **13** together, the controller **200** (or **50**) can be arranged to perform the process **800**. The controller **200** can receive an amplitude signal indicative of a real-time amplitude of ultrasonic vibration of the welding device **114**

during a treatment cycle (Step **810**). The received amplitude signal can be divided or segmented (for example, by the processor **210**) into a plurality of amplitude segments for the treatment cycle (Step **820**). The amplitude value of each amplitude segment of the plurality of amplitude segments can be monitored (for example, by the processor **210**) during the treatment cycle (Step **830**) and a plurality of real-time amplitude adjustment values determined or calculated by a plurality of closed-loop control algorithms, with each of the plurality of amplitude adjustment values corresponding to a respective one of the plurality of amplitude segments (Step **840**). The real-time amplitude adjustment value of each closed-loop control algorithm is based, in part, on historical amplitude response data. Each amplitude adjustment value can be applied to the corresponding respective one of the plurality of amplitude segments during the treatment cycle in real-time (Step **850**), thereby applying what is detected on a present bond to the next bond, or subsequent (or next) treatment. The process **800** when performed can reduce amplitude deviations prior to and/or during bonding operations.

FIG. **11B** illustrates an embodiment of a process **1000** of the welding device **100** equipped with ACAR, according to the principles of the disclosure. Referring to FIGS. **11B** and **13**, the process starts (**1002**) with an IDLE state in which the controller **200** waits for a bond status signal (**1004**). In a First Bond Detect state, the bond slice time duration is calculated by the controller **200**, and a bond slice timer is started (**1006**). The bond start time is sampled for bond period estimation by the controller **200** (**1006**). In a First Bond Slice N Controller state (where N is an integer, such as 1, 2, 3, and so on), the amplitude error is sampled into bond slice N amplitude via the controller **200**, and N is incremented (**1010**).

Unless N exceeds the number of slices (**1012**), the bond slice timer is started for slice N (**1014**) and the process **1000** returns to state **1008**. Otherwise, the process **1000** waits for the bond status signal to clear (**1016**), and when it does, the ACAR output is removed, and a pre-boost timer is started by the controller **200** to prepare for the next bond and wait until the timer elapses (**1018**).

When the pre-boost timer elapses, the pre-boost ACAR output is ramped by the controller **200**, and the amplitude signal is monitored for safe levels (**1020**). When bond start is detected by the controller **200**, the amplitude error is sampled for pre-boost, and N is set to 0 by the controller **200** (**1022**). Bond slice 0 is applied to the controller ACAR output, the bond slice time duration is calculated, and the bond slice timer is started by the controller **200** (**1022**).

The process **1000** transitions to the Bond Slice N state (**1024**) and the controller **200** waits for the bond slice timer to elapse (**1018**). Once the bond slice timer elapses, the process **1000** runs the bond slice N (**1026**), samples the amplitude error into bond slice N amplitude controller, and increments N via the controller **200** (**1026**). Unless N exceeds the number of slices (**1028**), the bond slice N controller ACAR output is applied, and the bond slice timer for slice N is started (**1030**), and the process **1000** returns to state **1024**. If N exceeds the number of slices (**1028**), the process **1000** moves to state **1016**.

Devices that are in communication with each other need not be in continuous communication with each other unless expressly specified otherwise. In addition, devices that are in communication with each other may communicate directly or indirectly through one or more intermediaries.

Although process steps, method steps, or algorithms may be described in a sequential or a parallel order, such pro-

cesses, methods and algorithms may be configured to work in alternate orders. In other words, any sequence or order of steps that may be described in a sequential order does not necessarily indicate a requirement that the steps be performed in that order; some steps may be performed simultaneously. Similarly, if a sequence or order of steps is described in a parallel (or simultaneous) order, such steps can be performed in a sequential order. The steps of the processes, methods or algorithms described in this specification may be performed in any order practical.

When a single device or article is described, it will be readily apparent that more than one device or article may be used in place of a single device or article. Similarly, where more than one device or article is described, it will be readily apparent that a single device or article may be used in place of the more than one device or article. The functionality or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality or features.

The terms “a,” “an,” and “the,” as used in this disclosure, means “one or more,” unless expressly specified otherwise.

The terms “communicating device” or “communication device,” as used in this disclosure, mean any computing device, hardware, or computing resource that can transmit or receive data packets, instruction signals or data signals over a communication link. The communicating device or communication device can be portable or stationary.

The terms “computer,” “computing device,” or “processor,” as used in this disclosure, means any machine, device, circuit, component, or module, or any system of machines, devices, circuits, components, or modules that are capable of manipulating data according to one or more instructions. The terms “computer,” “computing device” or “processor” can include, for example, without limitation, a processor, a microprocessor (μ C), a central processing unit (CPU), a graphic processing unit (GPU), a data processing unit (DPU), an application specific integrated circuit (ASIC), a general purpose computer, a super computer, a personal computer, a laptop computer, a palmtop computer, a notebook computer, a desktop computer, a workstation computer, a server, a server farm, a computer cloud, or an array or system of processors, μ Cs, CPUs, GPUs, ASICs, general purpose computers, super computers, personal computers, laptop computers, palmtop computers, notebook computers, desktop computers, workstation computers, or servers.

The term “computer-readable medium,” as used in this disclosure, can mean any non-transitory storage medium that participates in providing data (for example, instructions) that can be read by a computer. Such a medium can take many forms, including non-volatile media and volatile media. Non-volatile media can include, for example, optical or magnetic disks and other persistent memory. Volatile media can include dynamic random-access memory (DRAM). Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, punch cards, paper tape, any other physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, a carrier wave as described hereinafter, or any other medium from which a computer can read. The computer-readable medium can include a “cloud,” which can include a distribution of files across multiple (e.g., thousands of) memory caches on multiple (e.g., thousands of) computers.

Various forms of computer readable media can be involved in carrying sequences of instructions to a computer.

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For example, sequences of instruction (i) can be delivered from a RAM to a processor, (ii) can be carried over a wireless transmission medium, or (iii) can be formatted according to numerous formats, standards or protocols, including, for example, WiFi, WiMAX, IEEE 802.11, DECT, 0G, 1G, 2G, 3G, 4G, 5G, or 6G cellular standards, or Bluetooth.

The terms “including,” “having,” “comprising,” and variations thereof, as used in this disclosure, mean “including, but not limited to,” unless expressly specified otherwise.

The subject matter described above is provided by way of illustration only and should not be construed as limiting. Various modifications and changes can be made to the subject matter described herein without following the example embodiments and applications illustrated and described, and without departing from the true spirit and scope of the invention encompassed by the present disclosure, which is defined by the set of recitations in the following claims and by structures and functions or steps which are equivalent to these recitations.

What is claimed is:

1. An apparatus having two or more treatment modules positionable relative to each other, wherein at least one of the treatment modules includes a joining device for treating one or more workpieces during a treatment operation, the apparatus comprising:

a controller configured to
determine a force applied to at least one portion of each of the one or more workpieces during each bond in a plurality of bonds of the workpiece,
analyze the force to determine a force characteristic value for each bond,
calculate a force adjustment value based on the force characteristic value for at least one of the plurality of bonds, and
apply the force adjustment value to at least one of the treatment modules,

wherein the force characteristic value for each bond is determined based on, for a predetermined period of time, at least one of
a peak of force values,
an average of force values,
a median of force values,
a range of force values,
a variance of force values, and
a standard deviation of force values.

2. The apparatus in claim 1, wherein:

the treatment operation comprises at least one of a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation; and
applying the force adjustment value to the at least one of the treatment modules reduces bond strength variability.

3. The apparatus in claim 1, wherein the controller is configured for closed loop control.

4. The apparatus in claim 1, wherein the controller is further configured to:

receive a force signal from a sensor that is indicative of the force applied to the at least one portion of each of the one or more workpieces during each bond, wherein the force is determined based on the received force signal; and

adjust a position of one of the two or more treatment modules when the force adjustment value exceeds a predetermined threshold.

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5. The apparatus in claim 4, wherein the force characteristic value for each bond is determined based on, for the predetermined period of time, the force signal, including said at least one of the:

peak of force values in the force signal;
average of force values in the force signal;
median of force values in the force signal;
range of force values in the force signal;
variance of force values in the force signal; and
standard deviation of force values in the force signal.

6. The apparatus in claim 1, wherein the force adjustment value is calculated based on at least one of:

an average of the force characteristic values of two or more bonds;
a median of the force characteristic values of two or more bonds;
a range of the force characteristic values of two or more bonds;
a variance of the force characteristic values of two or more bonds; and
a standard deviation of the force characteristic values of two or more bonds.

7. The apparatus in claim 1, wherein the predetermined period of time comprises a bond duration.

8. The apparatus in claim 1, wherein the controller is further configured to:

receive a force signal from a sensor, wherein the force characteristic value for each bond is determined based on the force signal; and
determine a range of force characteristic values based on the force characteristic values for a plurality of bonds.

9. The apparatus in claim 1, wherein the joining device comprises an ultrasonic horn configured to apply ultrasonic energy to the workpiece.

10. An apparatus having two or more treatment modules positionable relative to each other, wherein at least one of the treatment modules includes a joining device for treating one or more workpieces during a treatment operation, the apparatus comprising:

a controller configured to:
determine a force applied to at least one portion of each of the one or more workpieces during each bond in a plurality of bonds of the workpiece;
analyze the force to determine a force characteristic value for each bond;
calculate a force adjustment value based on the force characteristic value for at least one of the plurality of bonds; and
apply the force adjustment value to at least one of the treatment modules,

wherein the controller is further configured to:

receive at least one of an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal; and
calculate at least one of a bond period and a bond duration based on the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal.

11. The apparatus in claim 1, wherein the controller is further configured to:

receive at least one of an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal; and
calculate the force characteristic value for each bond based on the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal.

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12. The apparatus in claim 1, wherein the controller is further configured to:

receive at least one of an ultrasonic amplitude signal, an ultrasonic power signal, and an ultrasonic frequency signal; and

calculate at least one of a bond period and a bond duration based on the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal.

13. The apparatus in claim 10, wherein the at least one of the ultrasonic amplitude signal, the ultrasonic power signal, and the ultrasonic frequency signal are received from an ultrasonic controller.

14. The apparatus in claim 1, wherein the controller is further configured to receive a position signal from a position sensor.

15. The apparatus in claim 14, wherein the position sensor comprises a linear encoder.

16. The apparatus in claim 6, wherein the range of force characteristic values is calculated as a difference between a minimum force characteristic value and a maximum force characteristic value amongst a plurality of force characteristic values, and wherein the range of force characteristic values correlates to bond quality amongst the plurality of bonds in the workpiece.

17. The apparatus in claim 16, wherein the controller is configured to generate an error signal based on the range of force characteristic values.

18. The apparatus in claim 1, wherein the workpiece includes one or more continuous webs or discrete pieces of material.

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19. A method for treating one or more workpieces during a treatment operation comprising two or more treatment modules positionable relative to each other, the method comprising:

5 determining a force applied to at least one portion of each of the one or more workpieces during each bond in a plurality of bonds of the workpiece;

analyzing the force to determine a force characteristic value for each bond;

10 calculating a force adjustment value based on the force characteristic value for at least one of the plurality of bonds; and

applying the force adjustment value to at least one of the treatment modules,

wherein the force characteristic value for each bond is determined based on, for a predetermined period of time, at least one of

a peak of force values,

an average of force values,

20 a median of force values,

a range of force values,

a variance of force values, and

a standard deviation of force values.

25 20. The method in claim 19, wherein the treatment operation comprises at least one of a bonding operation, a welding operation, a soldering operation, a fusing operation, or a cutting operation and applying the force adjustment value to the at least one of the treatment modules reduces bond strength variability.

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